AUTOMATION CASE STUDY

Make-ready sequencing

Vendor updates become verified stage changes and manager-approved handoffs.

A vacant unit can wait between trades because updates arrive in different places. I built a coordination automation that reads vendor replies, checks the work sequence, drafts the next handoff and gives the manager control over every vendor send.

SEVEN CONNECTED WORKFLOWS

Vendor intake, stage updates, manager controls, delivery, Airtable reconciliation, monitoring and error handling.

HUMAN RELEASE

Email, Message or Both is selected on an actual Telegram approval card. An ignored card remains open.

OBSERVABLE RESULTS

The sample connects vendor replies to saved executions, delivered messages and the resulting Airtable records.

DEMONSTRATION BOUNDARIES

Cedar Ridge, its vendors, units and addresses are fictional demo data. n8n and Postgres run locally; Telegram, Airtable, the test mailbox and OpenAI are real services used for the demonstration. No application screens are fabricated. The mailbox uses a single-inbox demo arrangement.

Execution evidence reuses preserved August 19-20 runs, with later error evidence identified separately. The application was inspected and screenshots captured in October. Canvas images show tidied editor drafts; execution images show the saved runs. This demonstrates coordination behavior, not measured customer savings.

ARCHITECTURE

How the systems connect

n8n orchestrates; Postgres is the authoritative record. Airtable is the manager-facing mirror and intake surface.

RECEIVE

Telegram, email or the generic inbound API creates one normalized message.

INTERPRET AND MATCH

OpenAI interprets wording. Assigned-vendor and thread rules identify the unit.

GUARD AND DRAFT

A shared sequence guard permits a completion and saves the next draft.

APPROVE AND DELIVER

A manager chooses the channel. Independent delivery records track the result.

DURABLE STATE

Postgres stores turns, stages, messages, events, approvals and one delivery row per authorized channel. Vendor and manual Done actions use the same guard. Ambiguous matches stop for a manager decision.

MANAGER VISIBILITY

Airtable mirrors operating records and accepts new-turn intake. Telegram carries approval cards, manual controls, daily digests and stall alerts. Mirror failures do not change sequencing state; unexpected errors have a separate builder route.

AI can classify a reply or rewrite fact-locked wording. It cannot approve a send, choose the unit, override the sequence or judge workmanship. A factual draft is available when AI drafting fails.

CONNECTED DEMONSTRATION

Follow one unit through three trades

Unit 3 / 2201 Oak Ave / fictional Cedar Ridge portfolio

01 / TRASH-OUT REPLY

QuickHaul replies in Telegram to the unit-specific chase. The router identifies Unit 3 and a completion intent.

02 / REPAIRS HANDOFF

Trash-out closes and Approval 87 is saved. The manager chooses Email; one email delivery is recorded as sent.

03 / PAINTING HANDOFF

The repairs email reply resolves to the same unit. Approval 88 is released as Message and arrives in Rivera Painting’s chat.

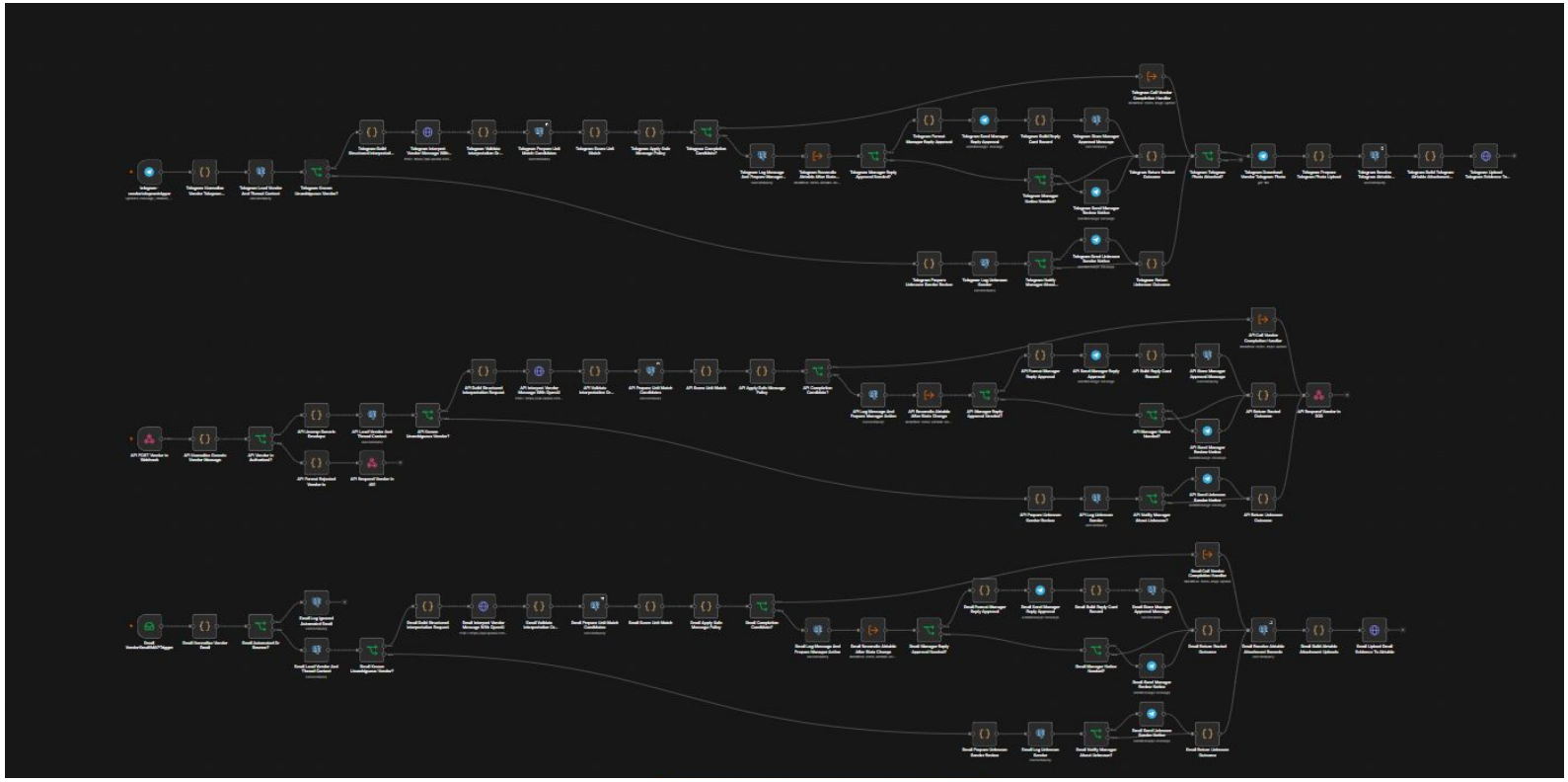
04 / NEXT APPROVAL REMAINS OPEN

The painting reply closes Paint. Approval 89 drafts Flooring, while Airtable shows Unit 3 awaiting approval.

WORKFLOW / VENDOR COMMUNICATIONS

Vendor Communications

Three entry routes converge on the same matching and message safety policy.



COMPLETE CANVAS / 97 NODES / NATIVE ZOOM CLOSE-UPS FOLLOW

TRIGGER / INPUT

Vendor Telegram post, mailbox reply or authenticated API payload.

STEPS / DECISIONS

Normalize, interpret, identify the assigned vendor and match the unit.

DATA / OUTPUT

Log the message; call Stage Update only for an accepted completion.

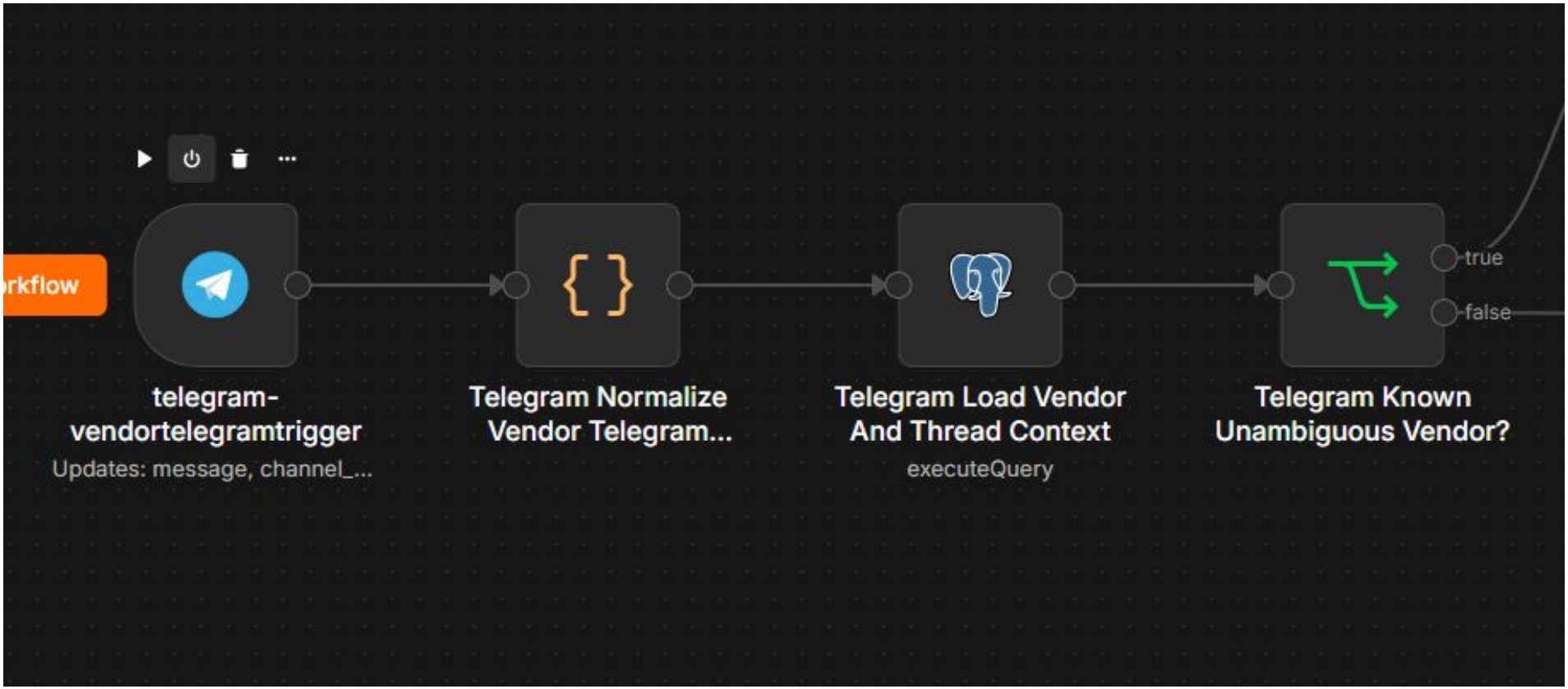
HUMAN CONTROL

Questions, ambiguity and reply drafts are routed for manager review.

VENDOR COMMUNICATIONS / CANVAS CLOSE-UP

Telegram intake

Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.

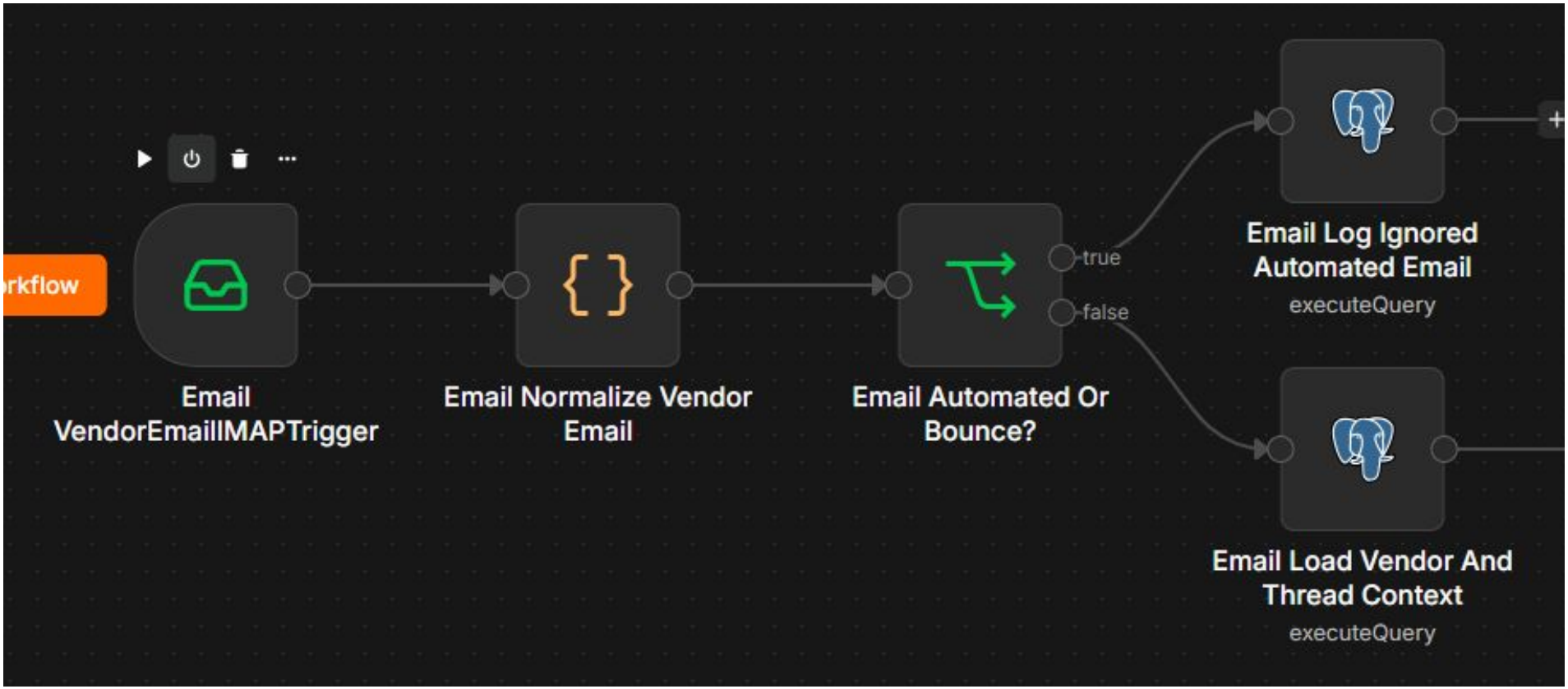


The vendor bot receives text and media. The message is normalized, then sender and reply-thread context are loaded from Postgres before interpretation.

VENDOR COMMUNICATIONS / CANVAS CLOSE-UP

Email intake

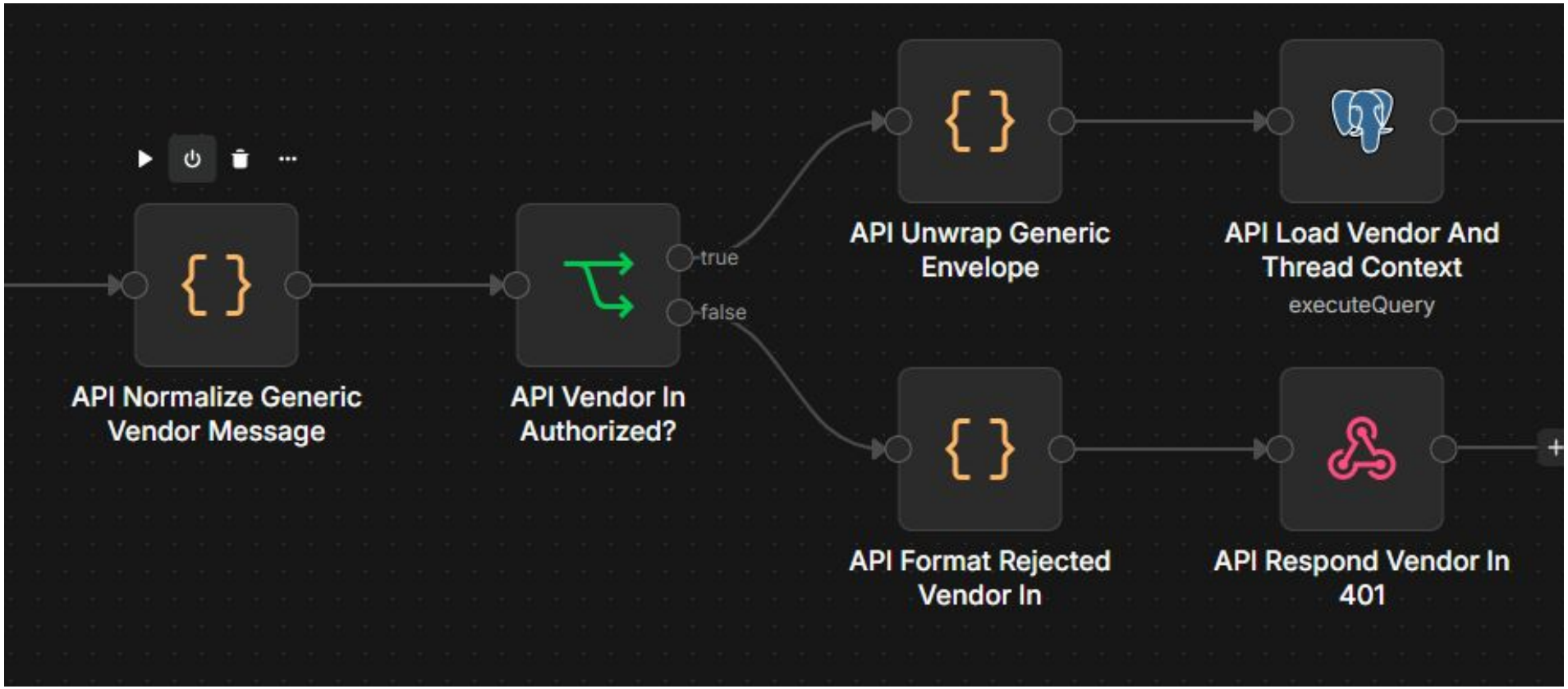
Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.



The mailbox route checks sender identity and message headers. Reply context or a reference token is resolved before sender/unit fallback. The demo ignores self-originated messages without reply context.

Generic inbound API

Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.

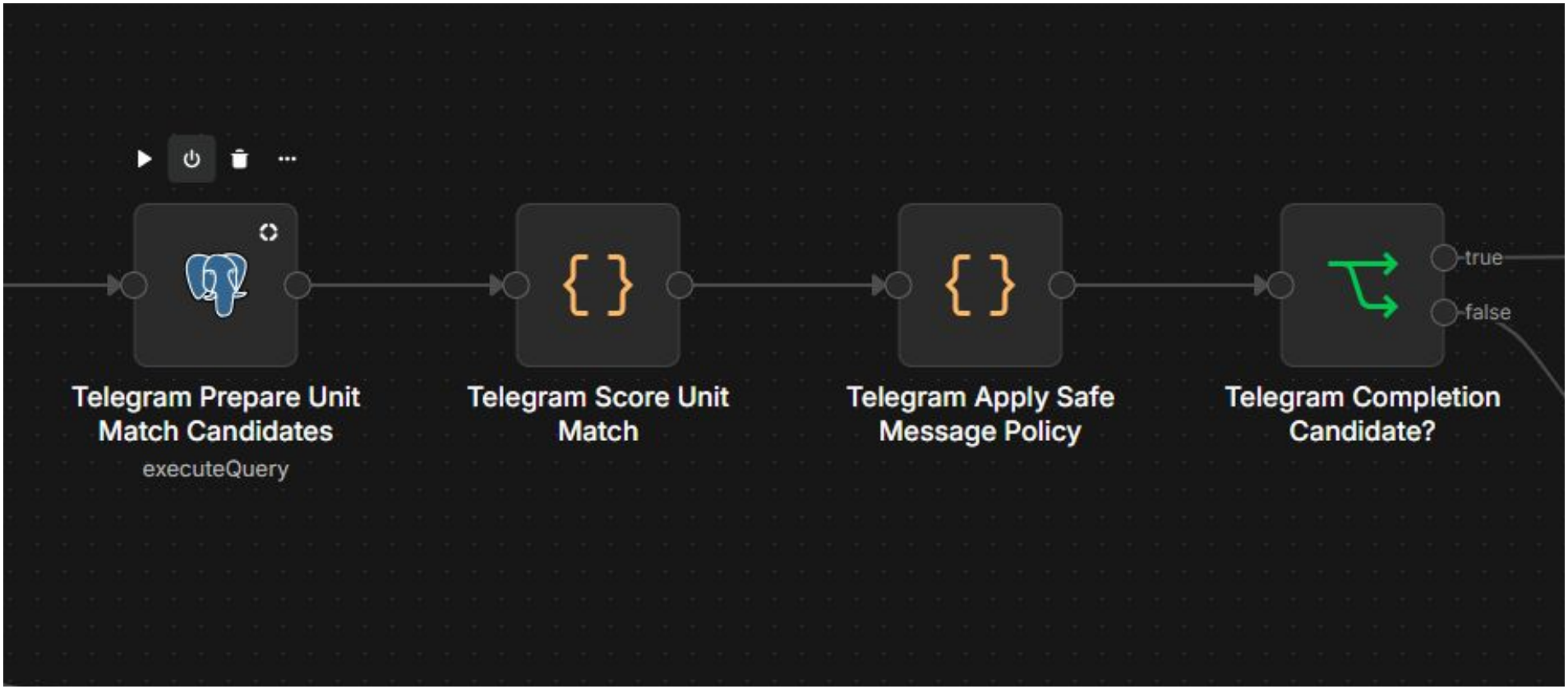


An authenticated inbound payload creates the same message envelope. Rejected requests exit through a separate response; accepted requests enter the common vendor policy.

VENDOR COMMUNICATIONS / CANVAS CLOSE-UP

Match the right unit

Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.

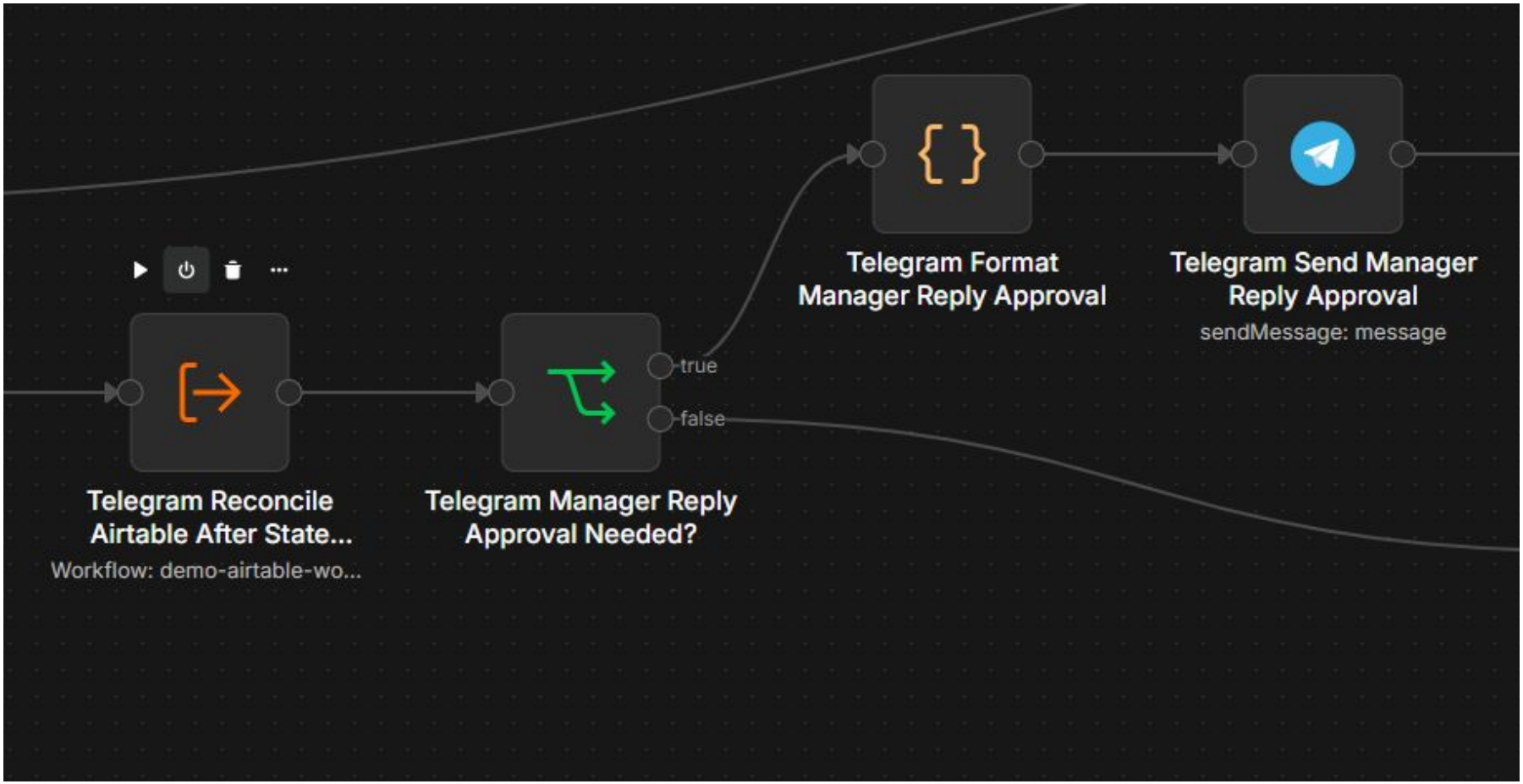


The model output is validated, assigned-unit candidates are prepared and a deterministic score selects a match. The safe-message policy decides whether a completion may proceed.

VENDOR COMMUNICATIONS / CANVAS CLOSE-UP

Review and reply approval

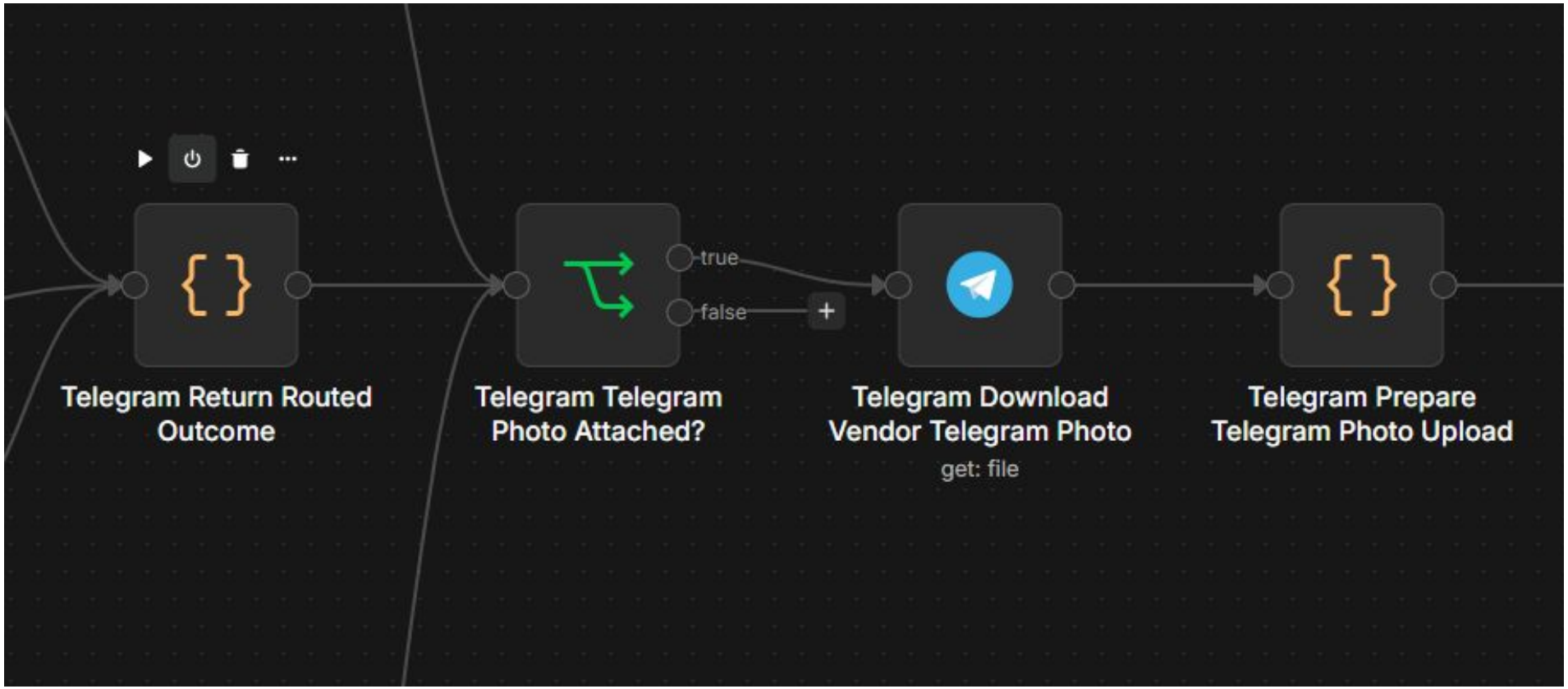
Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.



A question or uncertain message is logged with context. A reply draft can become a manager approval; the workflow does not contact a vendor by itself.

Photos and durable records

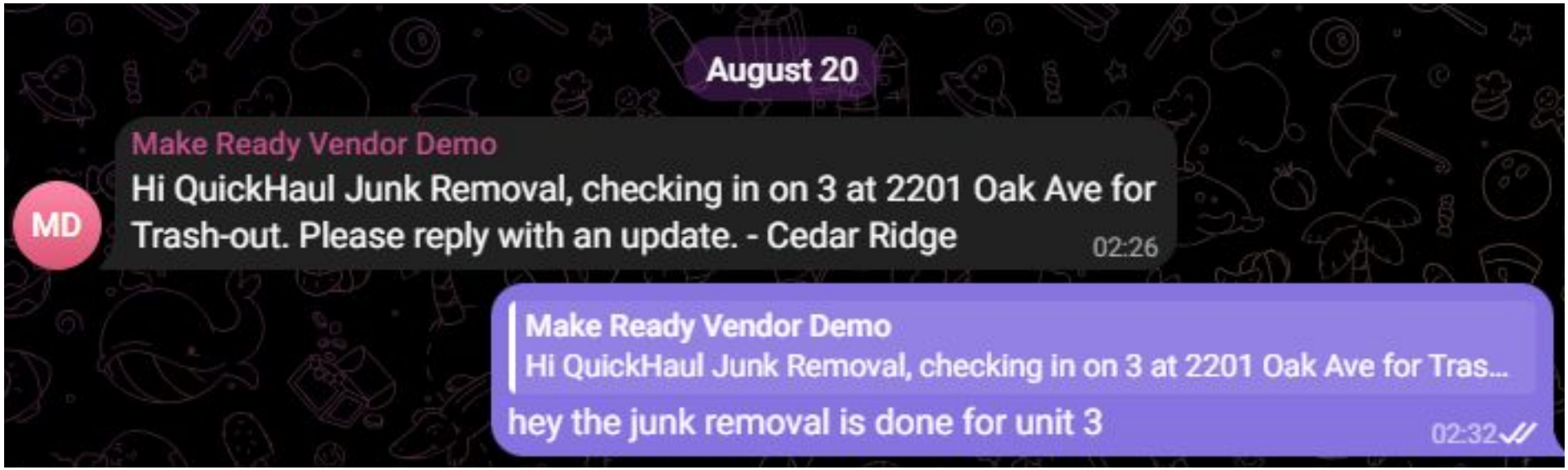
Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.



Attached Telegram photos are fetched and their receipt recorded. The photo is evidence of a submitted attachment; the automation does not certify workmanship.

The input: a reply in context

Telegram conversation / fictional vendor and unit / preserved August 20 demonstration



QuickHaul's actual demo conversation contains the unit-specific chase and the reply "hey the junk removal is done for unit 3." The reply context connects this input to the subsequent Unit 3 completion.

Interpret the wording

Actual application evidence from the synthetic demonstration.



```
"interpretation": {
  "model": "gpt-5.6-luna",
  "intent": "stage_complete",
  "status": "openai",
  "eta_text": null,
  "unit_hint": "3",
  "confidence": 0.99,
  "stage_hint": "Trash-out",
  "reply_needed": true,
  "safety_flags": [
  ],
  "manager_summary": "QuickHaul Junk Removal reports that the junk removal for unit 3 is
                      complete.",
  "suggested_reply": "Thanks for the update. We've noted the trash-out for unit 3 as
                      complete.",
  "ambiguity_reason": null
},
```

Saved execution 2691 classified the reply as stage_complete, with Trash-out as the stage hint. The live model result is interpretation input; the deterministic unit matcher and sequence guard still decide what may change.

Resolve the unit from its thread

Actual application evidence from the synthetic demonstration.




```
"match": {
  "selected": {
    "turn_id": 942,
    "stage_id": 10378,
    "unit_label": "3",
    "property_address": "2201 Oak Ave",
    "stage_code": "TRASHOUT",
    "stage_name": "Trash-out",
    "assigned_vendor_id": 934,
    "score": 100,
    "reasons": [
      "reply_thread"
    ]
  },
  "score": 100,
  "needs_manager": false,
```


The same execution selected turn 942 / Unit 3 and its Trash-out stage. The reason is reply_thread, with a deterministic match score of 100. This avoids guessing from informal wording alone.

VENDOR COMMUNICATIONS / OBSERVED RESULT

An ambiguous reply stops for review

Actual application evidence from the synthetic demonstration.



In saved execution 2681, “this task is done” could refer to the painter’s Unit 12B or 14A. The result is manager_match_needed, with no queue or draft. This preserved run records the review decision without sending a manager notification; it does not claim a delivered alert.

WORKFLOW / STAGE UPDATE

Stage Update

One shared completion handler protects the trade sequence and prepares the next handoff.



COMPLETE CANVAS / 20 NODES / NATIVE ZOOM CLOSE-UPS FOLLOW

TRIGGER / INPUT

A safe vendor completion or manager Done action with unit/stage context.

STEPS / DECISIONS

Reload state and apply prerequisite and repeat-completion checks.

DATA / OUTPUT

Complete the stage; save the next draft; reconcile the mirror.

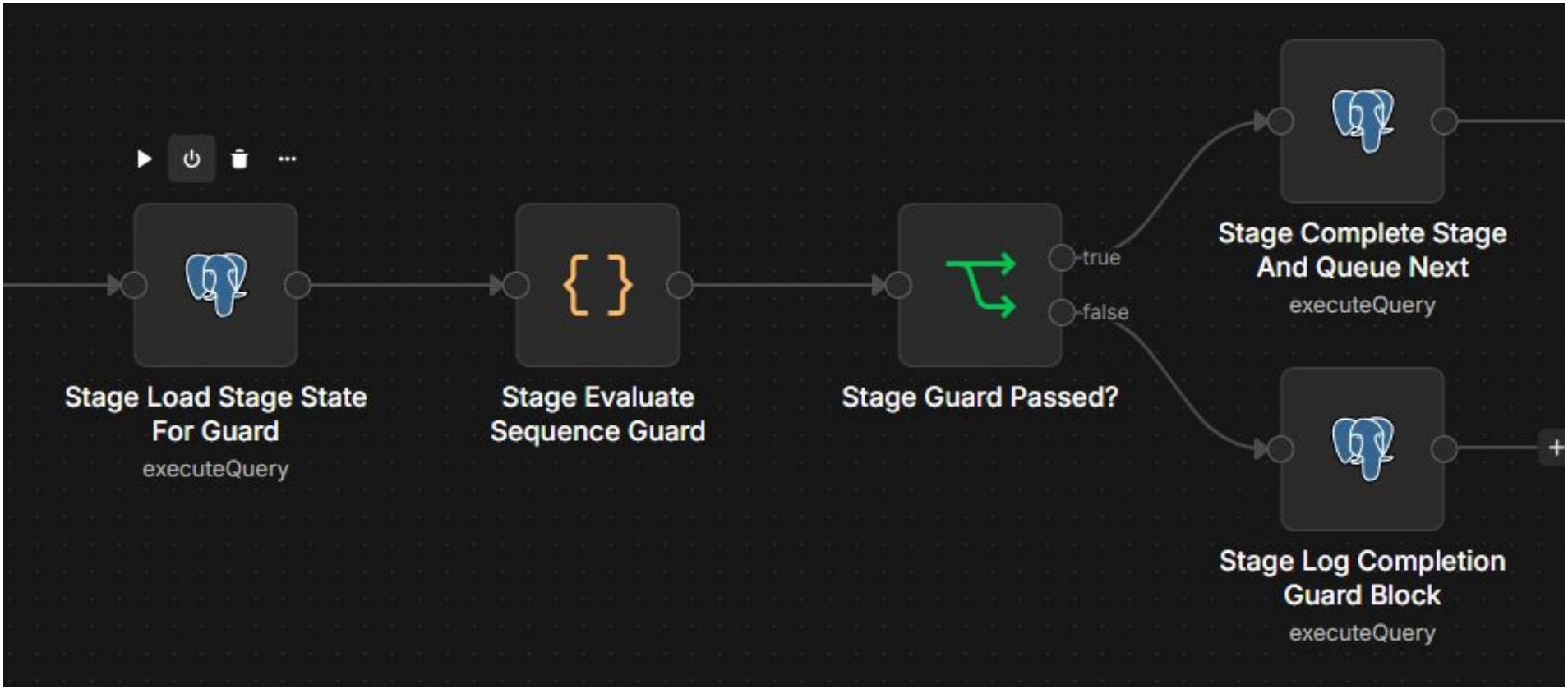
HUMAN CONTROL

Send an approval card. The draft is still unsent until a channel is chosen.

STAGE UPDATE / CANVAS CLOSE-UP

One deterministic sequence guard

Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.

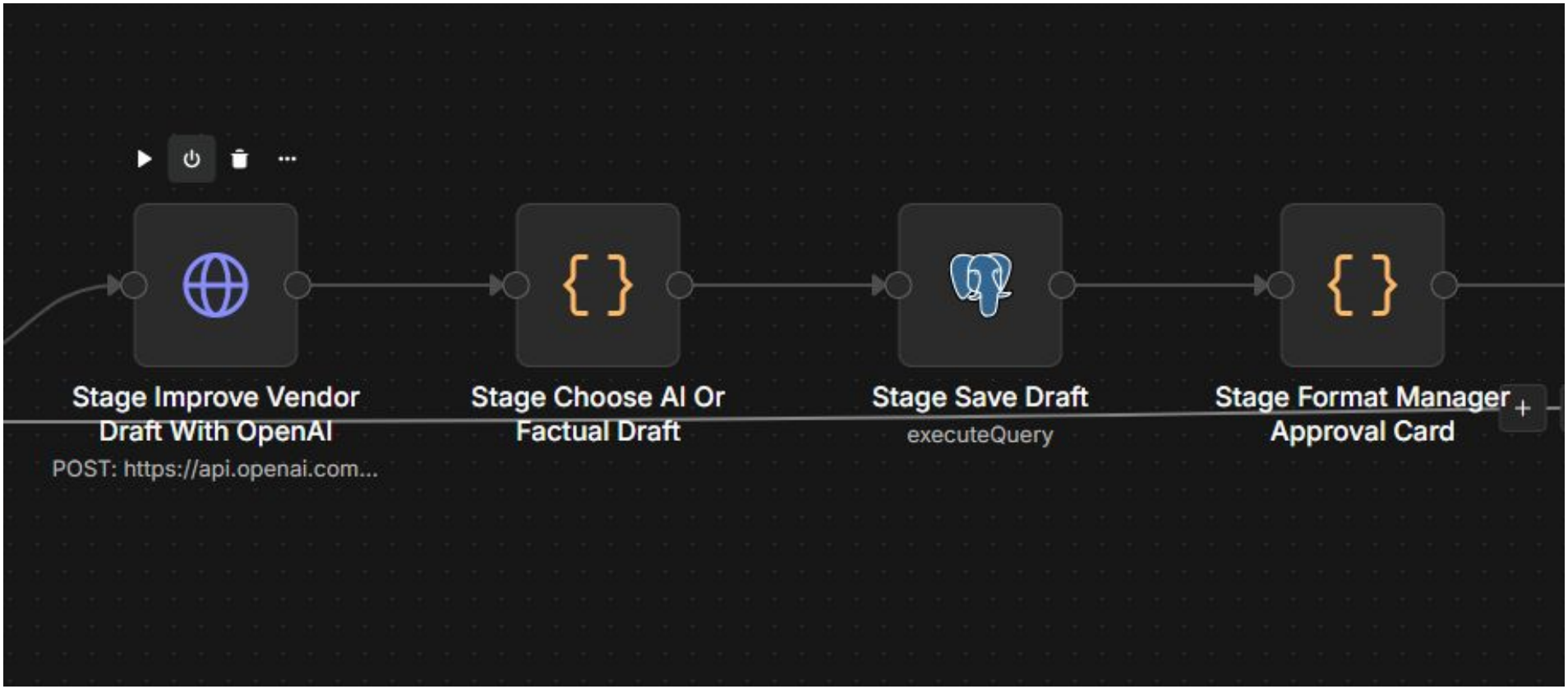


The guard checks the current stage and its prerequisites. An allowed completion takes the upper branch; a blocked action is logged on the lower branch. Manager and vendor completions share this check.

STAGE UPDATE / CANVAS CLOSE-UP

Build a fact-locked next draft

Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.

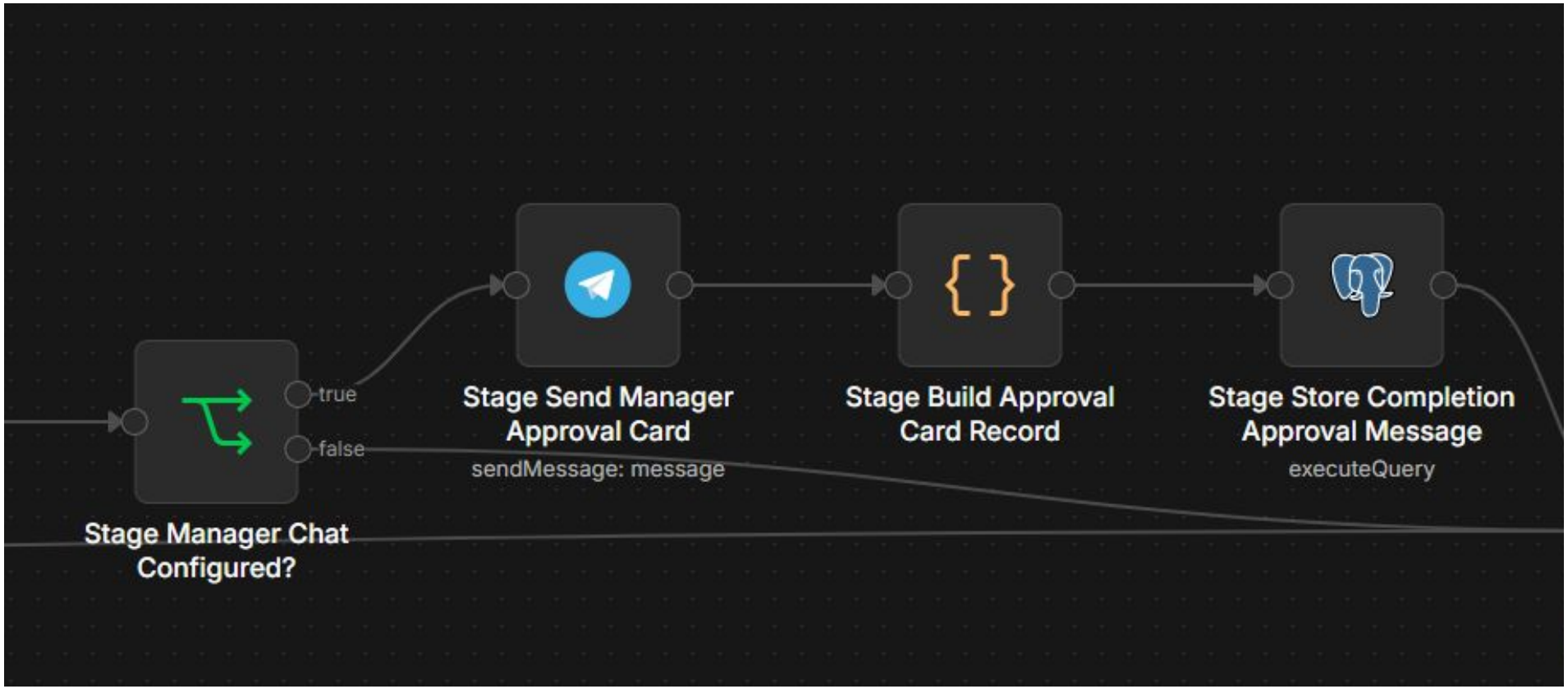


The next assigned vendor and stage facts are loaded from durable state. AI wording is optional; the workflow chooses AI or factual fallback, then saves the draft before requesting approval.

STAGE UPDATE / CANVAS CLOSE-UP

Create the manager approval

Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.

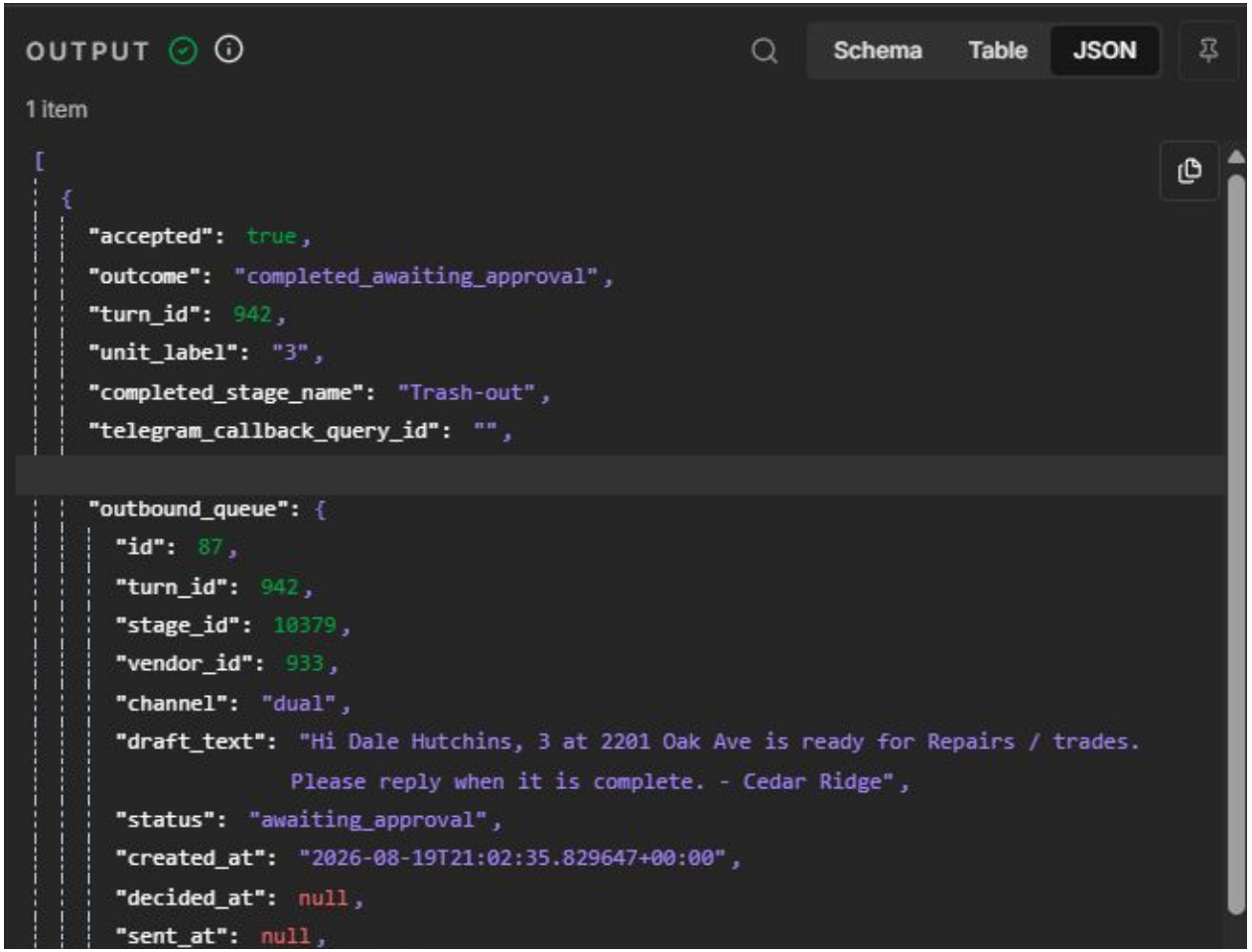


The saved draft becomes a Telegram approval card when the manager chat is configured. The approval message is logged and a completion outcome returns to the caller. This branch sends to the manager, not the vendor.

STAGE UPDATE / OBSERVED RESULT

Trash-out complete; repairs draft saved

Actual application evidence from the synthetic demonstration.



Execution 2692 accepted Unit 3’s Trash-out completion and returned completed_awaiting_approval. Approval 87 contains the factual repairs handoff to Dale Hutchins. The queue is awaiting approval, with no decision or sent time.

STAGE UPDATE / OBSERVED RESULT

A repeated completion is blocked

Actual application evidence from the synthetic demonstration.

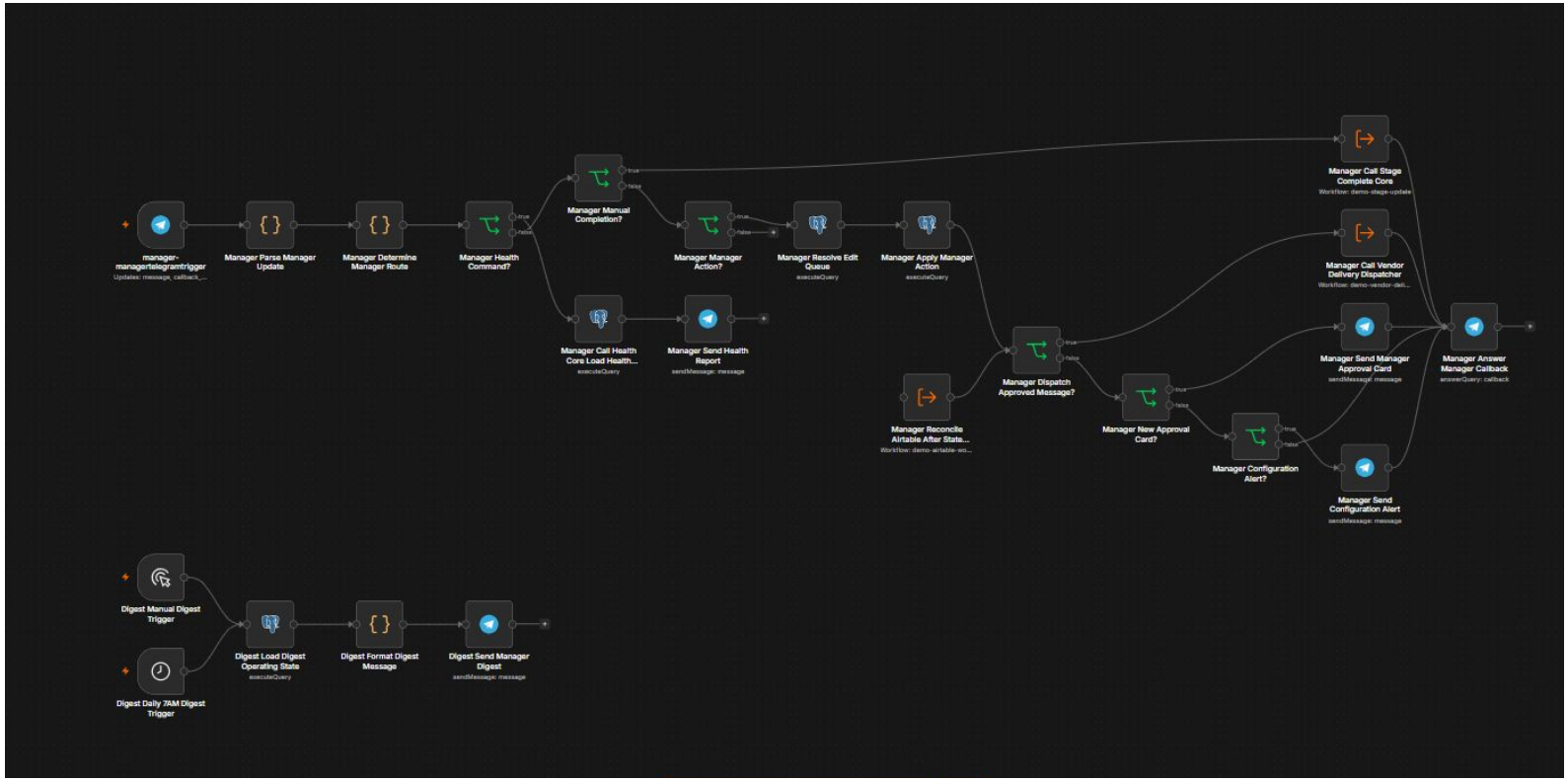


A separate preserved manager Done run (2169) returns allowed: false because Deep clean is already complete. QA remains active in the loaded state. The shared guard prevents this repeated request from advancing the unit again.

WORKFLOW / MANAGER OPERATIONS

Manager Operations

The manager makes operational decisions in Telegram, including every outbound channel choice.



COMPLETE CANVAS / 24 NODES / NATIVE ZOOM CLOSE-UPS FOLLOW

TRIGGER / INPUT

Manager bot message or callback; scheduled morning digest.

STEPS / DECISIONS

Parse commands; route health, Done, approval and other controls.

DATA / OUTPUT

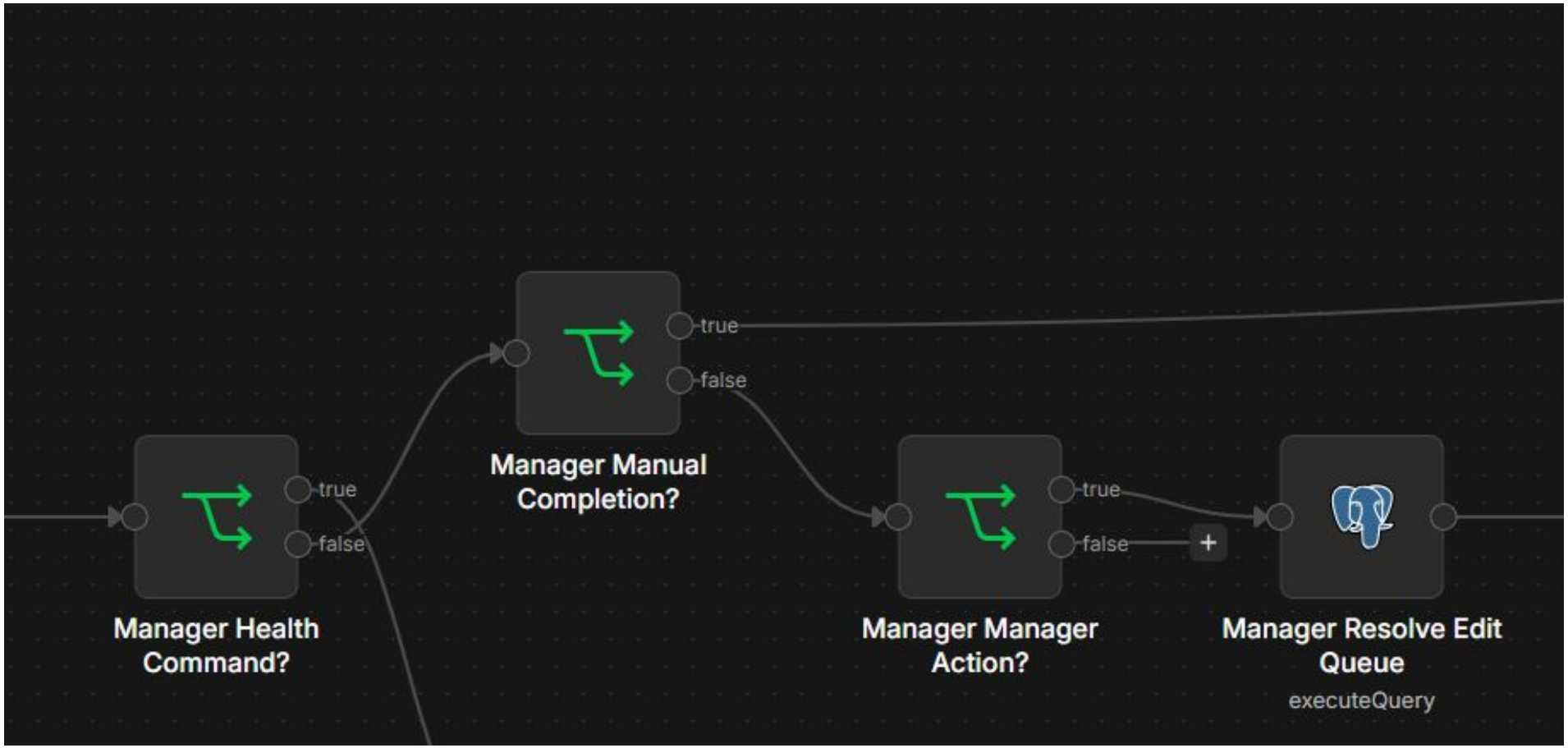
Persist the action; dispatch an approved queue; acknowledge the callback.

HUMAN CONTROL

Email, Message or Both releases a draft. Manual Done uses the shared guard.

Route the manager’s action

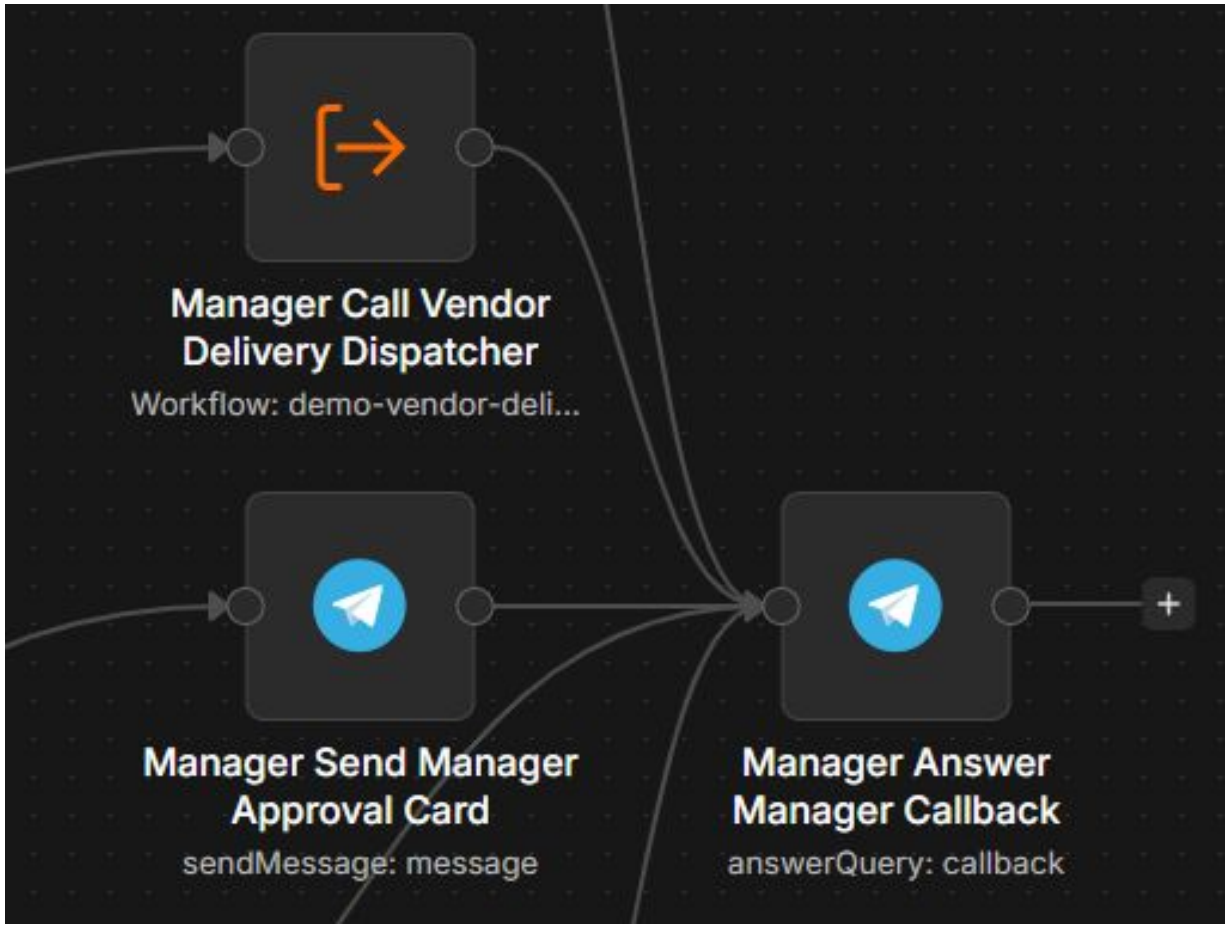
Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.



A health request returns operating metrics. A manual completion calls Stage Update. Other actions load the appropriate queue and persist the manager’s decision.

Dispatch only after approval

Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.

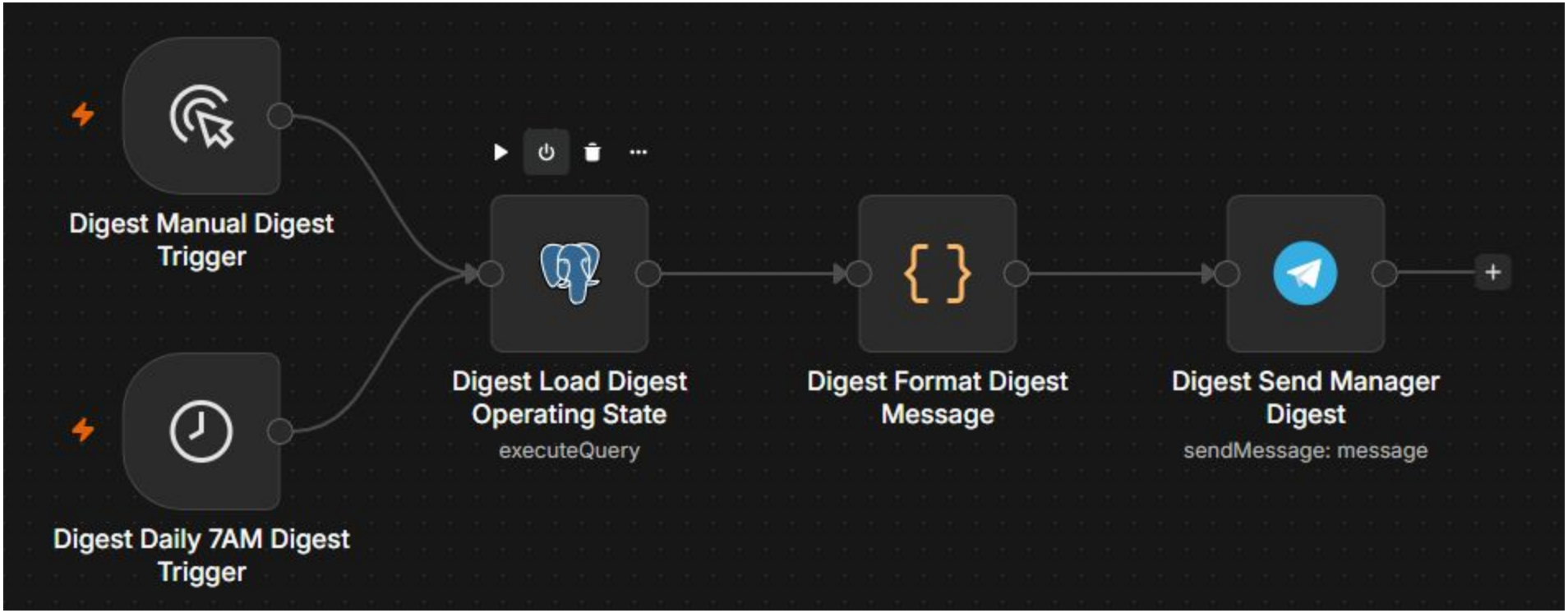


The approved action is recorded before the dispatch branch calls Vendor Delivery. Other branches show a new approval card or a configuration alert, then acknowledge the manager callback.

MANAGER OPERATIONS / CANVAS CLOSE-UP

Daily operating digest

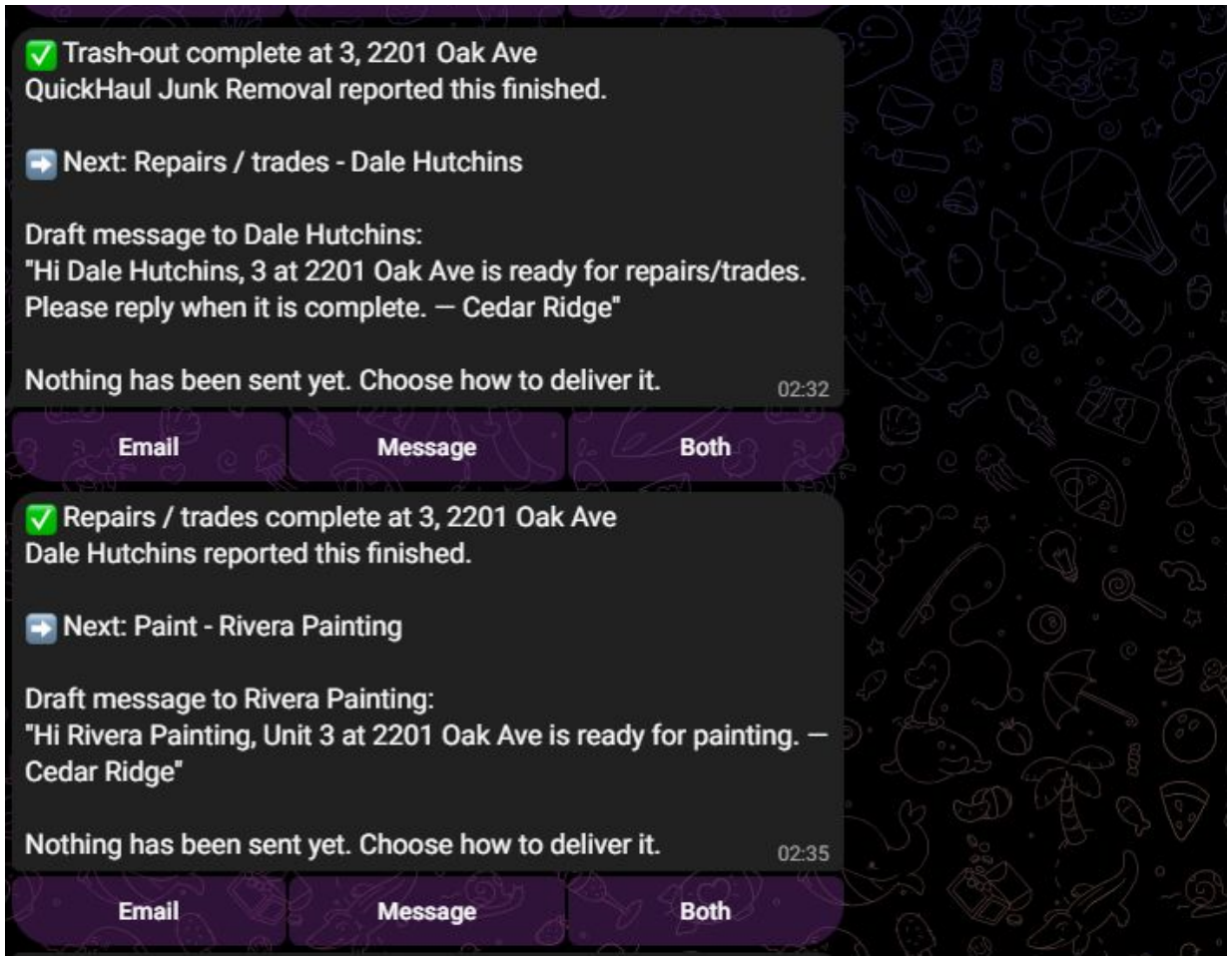
Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.



A manual trigger or the 7 AM schedule loads Postgres operating state, formats the digest and sends it to the manager bot. The manager receives a summary and controls in the same operating channel.

The handoffs remain visible in chat

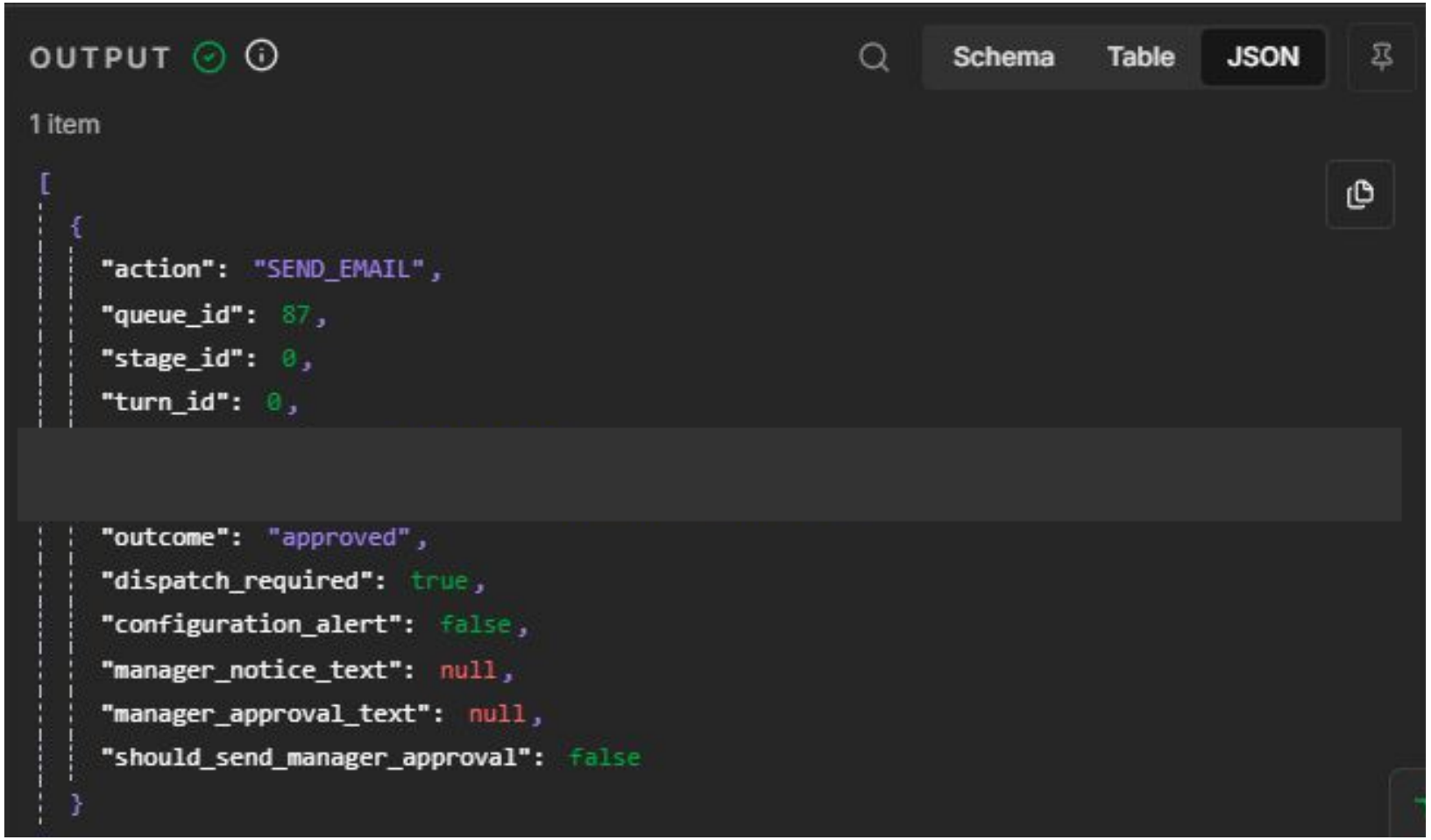
Actual manager Telegram conversation / fictional portfolio records



The active manager conversation shows Trash-out complete → Repairs and Repairs complete → Paint for Unit 3. Each card offers Email, Message and Both, and states that nothing has been sent yet. Keeping the conversation context shows progression between trades.

The human chose Email

Actual application evidence from the synthetic demonstration.

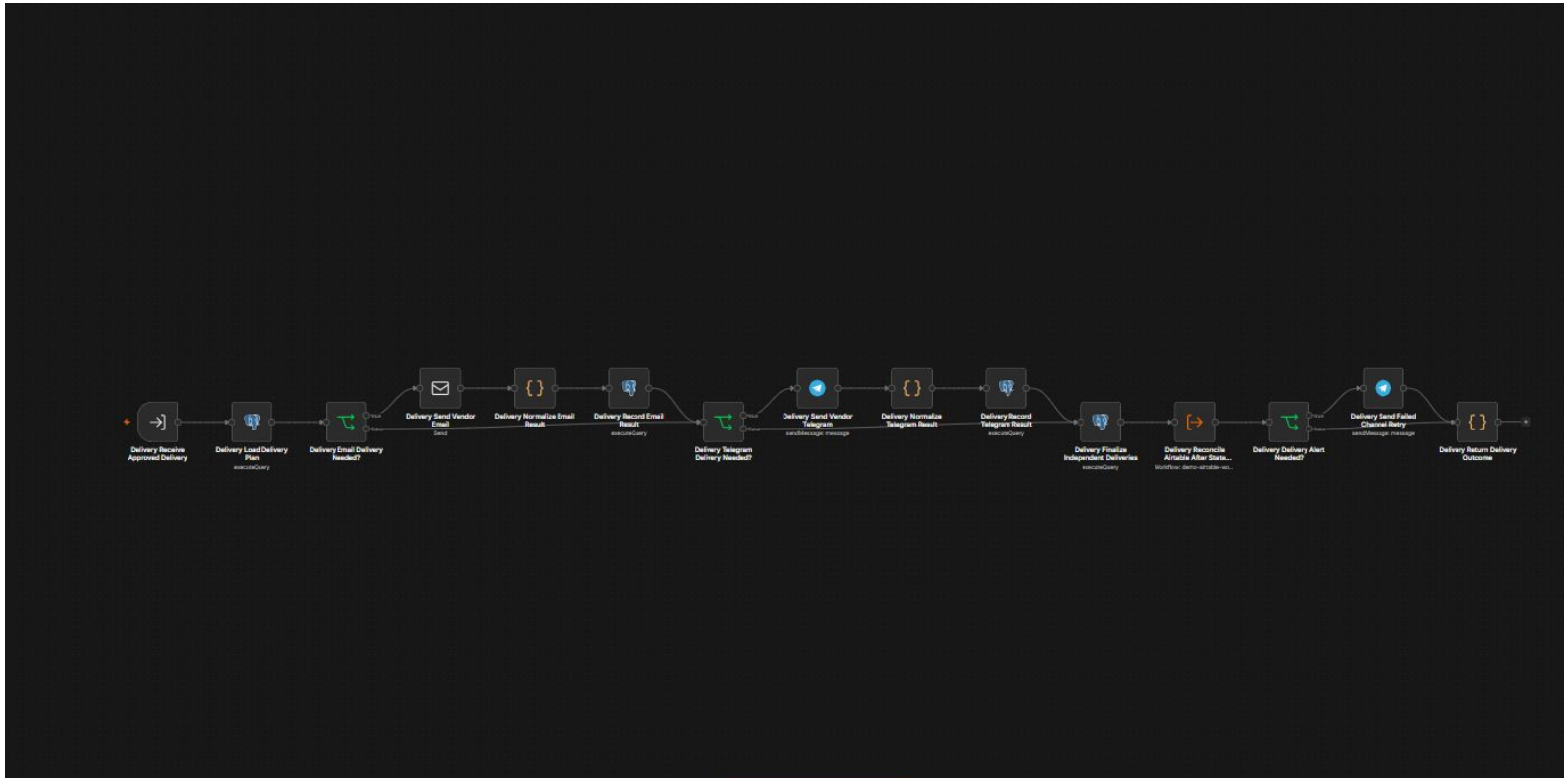


Saved manager execution 2698 records SEND_EMAIL for Approval 87, outcome approved and dispatch_required: true. This is the release point for the repairs handoff; the subsequent delivery result proves what happened next.

WORKFLOW / VENDOR DELIVERY

Vendor Delivery

A manager-approved queue becomes independent delivery attempts for the authorized channels.



COMPLETE CANVAS / 15 NODES / NATIVE ZOOM CLOSE-UPS FOLLOW

TRIGGER / INPUT

Called with an approved queue. It has no autonomous outbound trigger.

STEPS / DECISIONS

Check destinations; send selected channels; record each provider result.

DATA / OUTPUT

Finalize against the channels actually authorized; activate after a success.

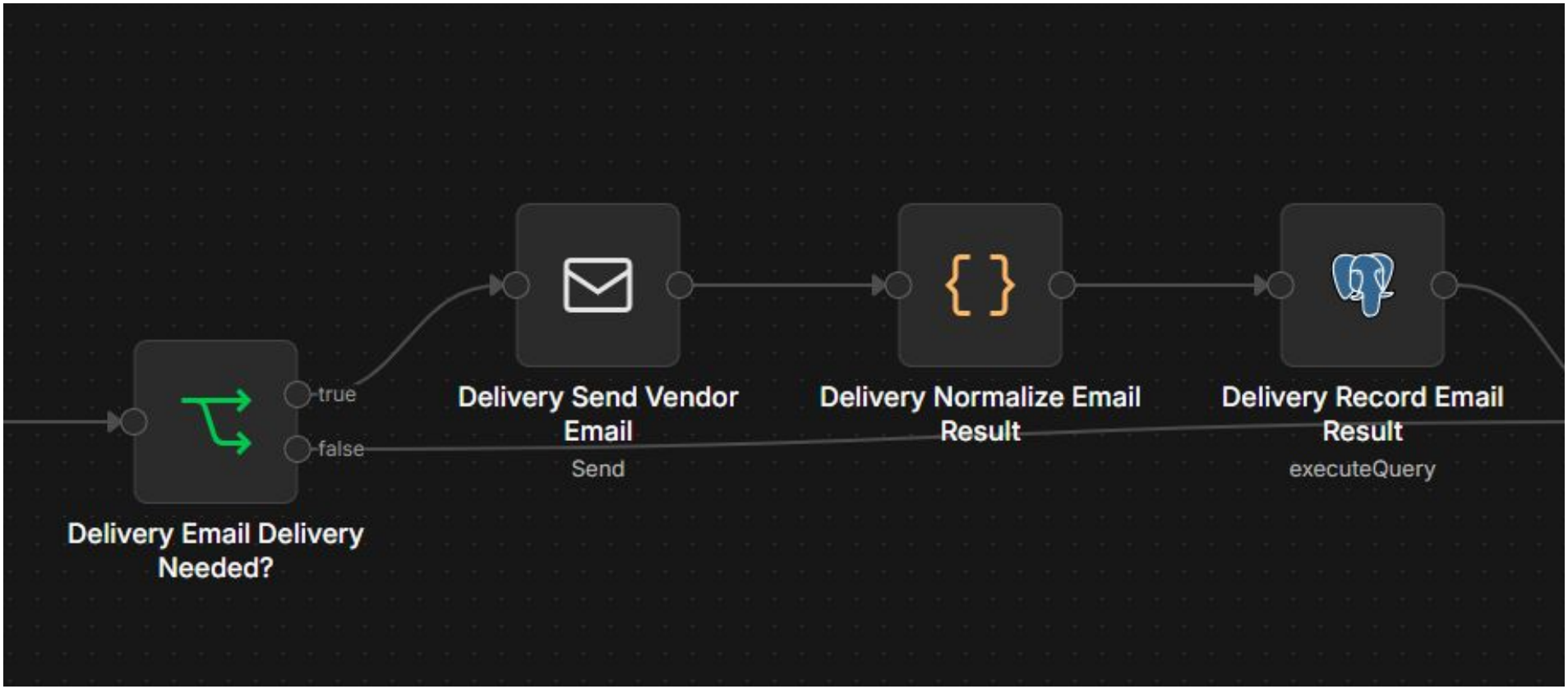
HUMAN CONTROL

Missing destination blocks that route; retry only a failed channel.

VENDOR DELIVERY / CANVAS CLOSE-UP

Email delivery and its result

Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.

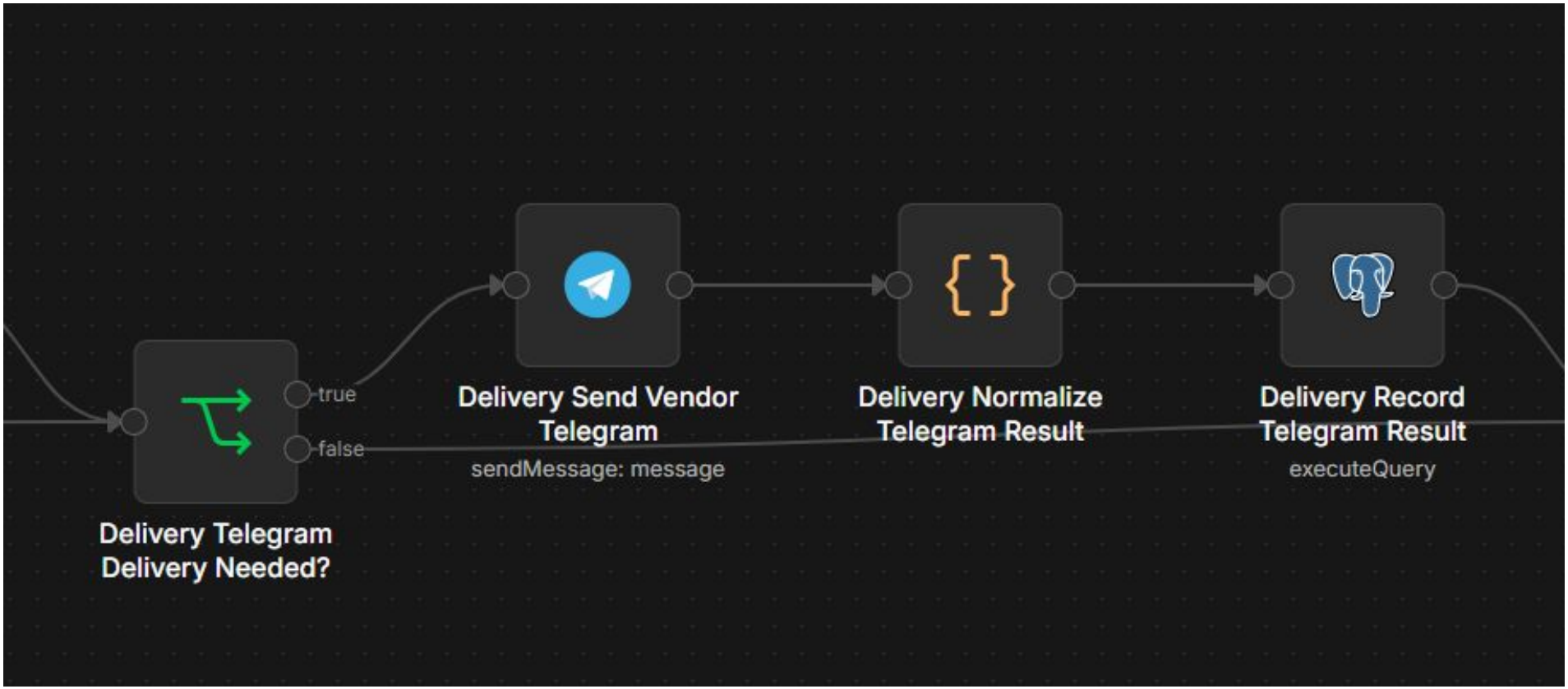


An Email authorization loads the current destination and attempts SMTP delivery. The result is normalized and recorded in its own durable row before the Telegram branch is considered.

VENDOR DELIVERY / CANVAS CLOSE-UP

Telegram is an independent route

Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.

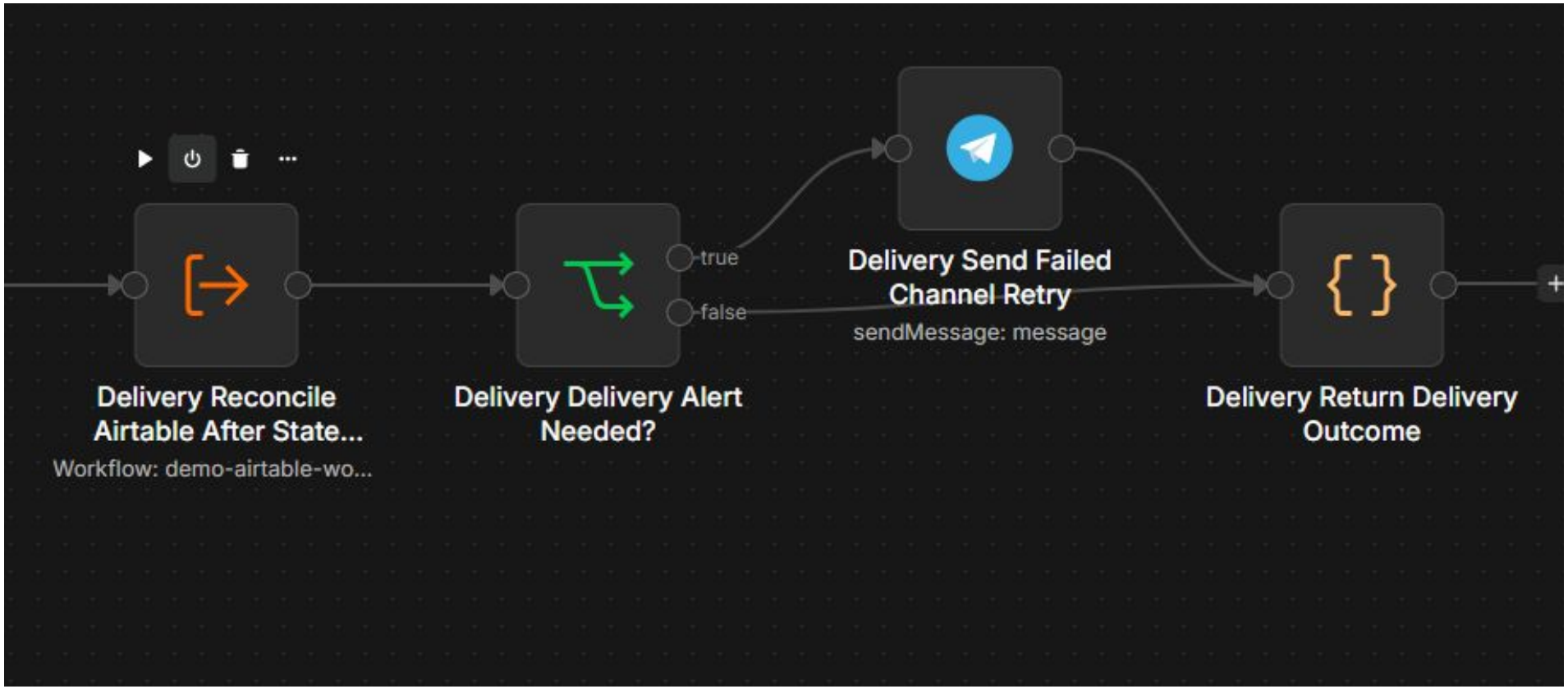


Message authorization attempts a Telegram send and stores its own provider result. Both authorizes two separate deliveries; one successful route is never inferred to prove the other succeeded.

VENDOR DELIVERY / CANVAS CLOSE-UP

Finalize, reconcile and alert

Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.

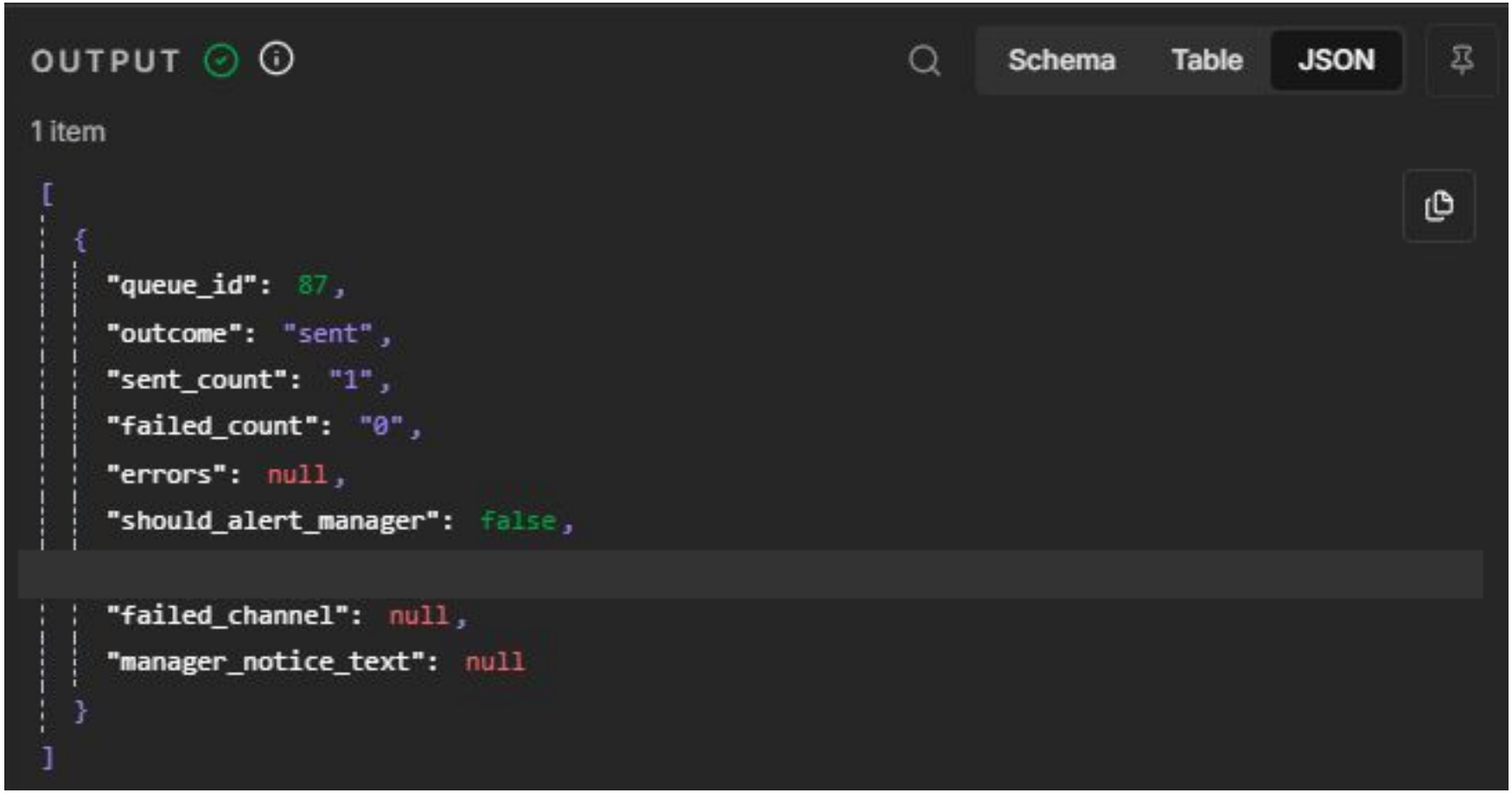


Finalization counts successful deliveries against the authorized channel count. The mirror is reconciled and a failure or configuration problem can notify the manager. Retry targets only the failed route.

VENDOR DELIVERY / OBSERVED RESULT

One authorized email, one sent result

Actual application evidence from the synthetic demonstration.

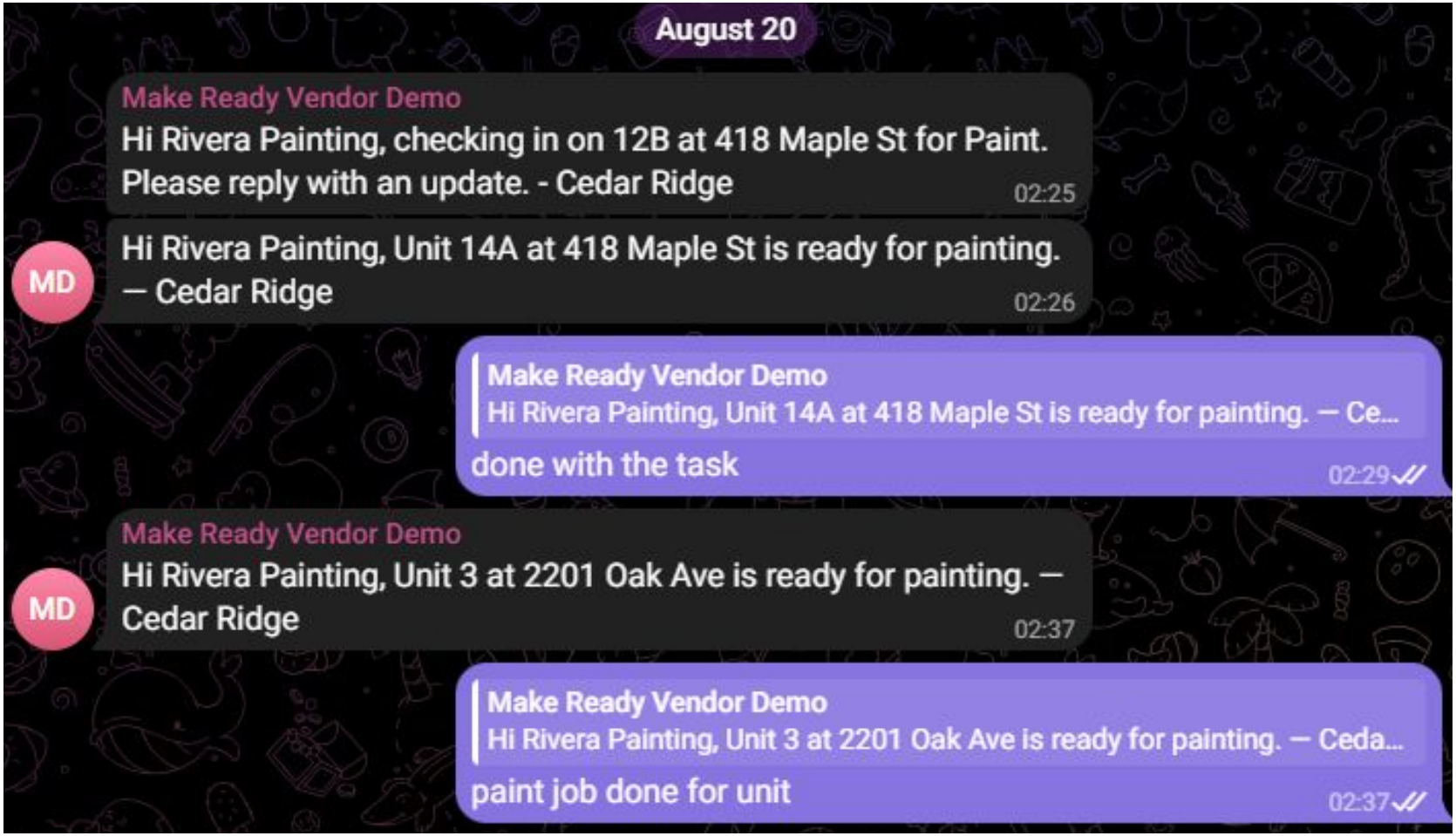


Execution 2699 finalized Approval 87 as sent: one successful delivery, zero failures. The manager had selected Email, so one success is sufficient. This is paired with the actual Airtable Email Sent record in the next section.

VENDOR DELIVERY / OBSERVED RESULT

The approved painting message arrived

Real Telegram delivery and vendor reply / synthetic names, units and addresses

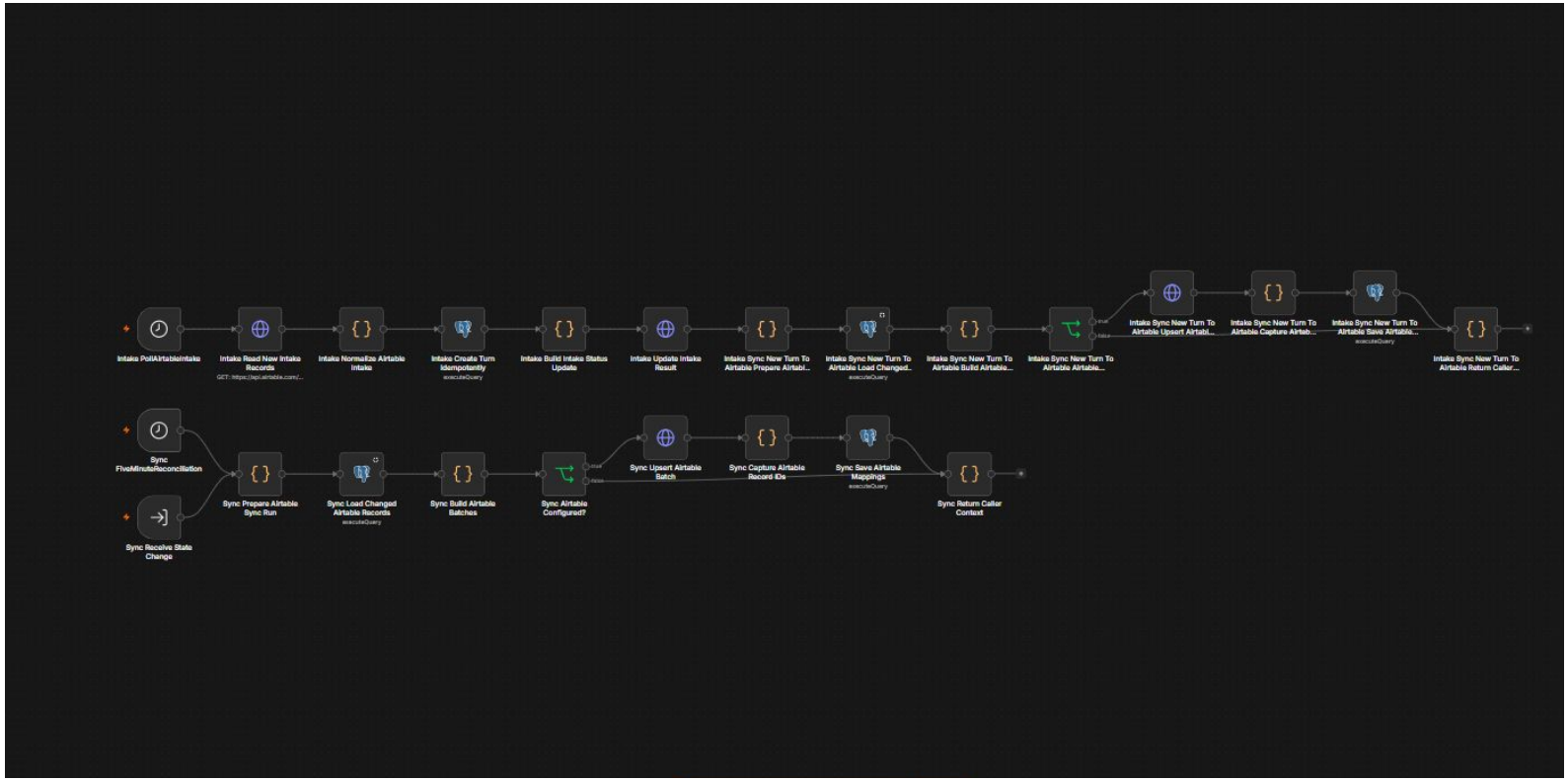


The actual Rivera Painting conversation shows the approved Unit 3 painting handoff and the vendor’s quoted reply “paint job done for unit.” Earlier Unit 12B and 14A messages remain in view, demonstrating why thread context matters for a vendor assigned to several units.

WORKFLOW / AIRTABLE WORKSPACE

Airtable Workspace

Intake creates durable turns; reconciliation mirrors coordination state for managers.



COMPLETE CANVAS / 24 NODES / NATIVE ZOOM CLOSE-UPS FOLLOW

TRIGGER / INPUT

New Airtable intake record; timer or state-change reconciliation call.

STEPS / DECISIONS

Validate intake; create idempotently; identify changed payloads.

DATA / OUTPUT

Batch typed records and retain external mappings for later sync.

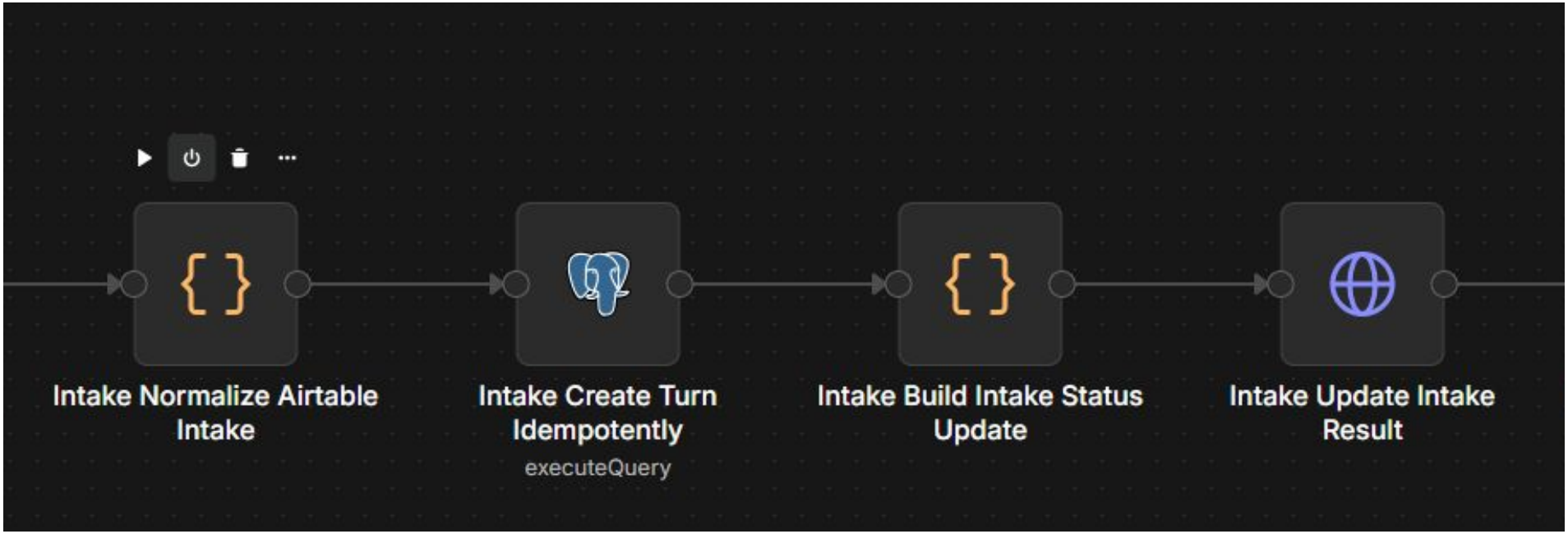
HUMAN CONTROL

Manager intake is the write surface. Postgres remains authoritative.

AIRTABLE WORKSPACE / CANVAS CLOSE-UP

Create a turn from manager intake

Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.

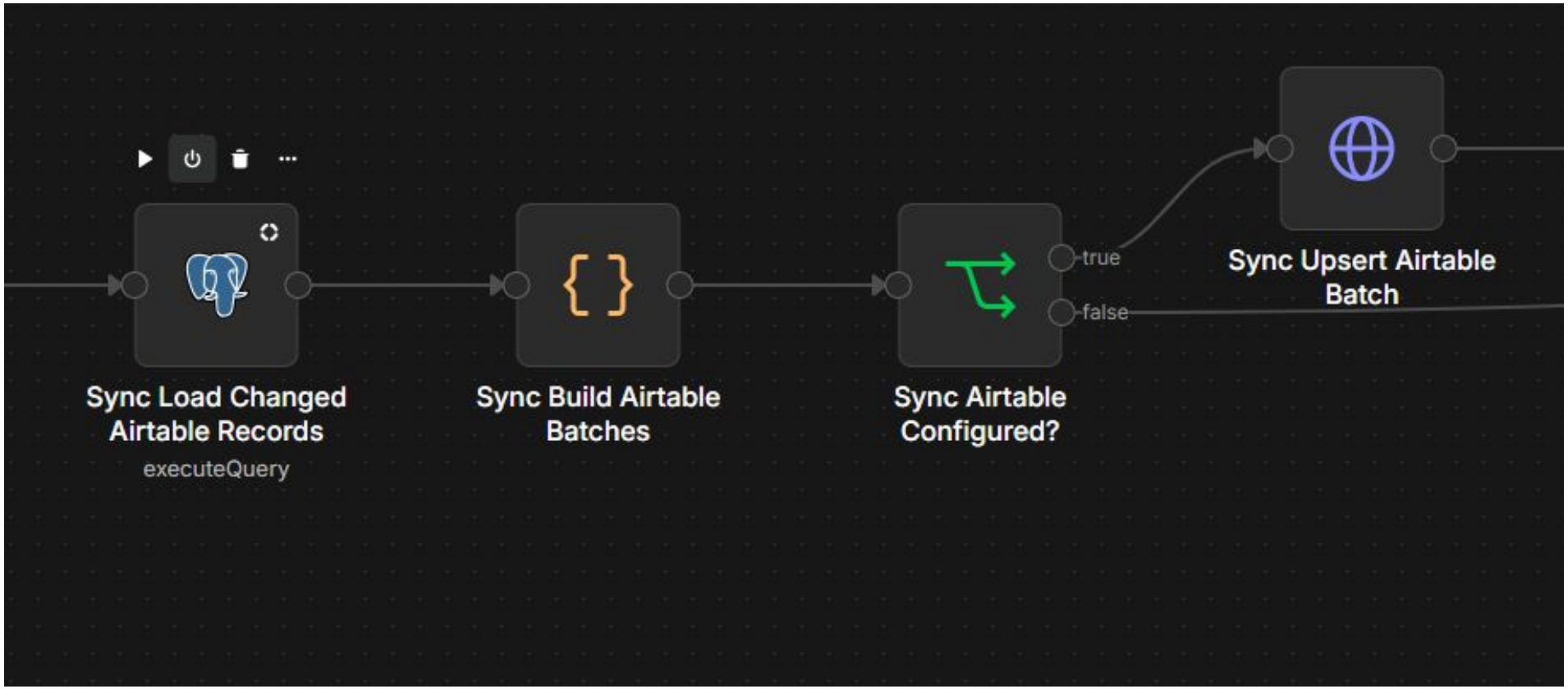


The poll reads new intake records, normalizes their fields and creates the turn and stages idempotently in Postgres. The intake record receives a result, then reconciliation updates the operating mirror.

AIRTABLE WORKSPACE / CANVAS CLOSE-UP

Batch the changed state

Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.

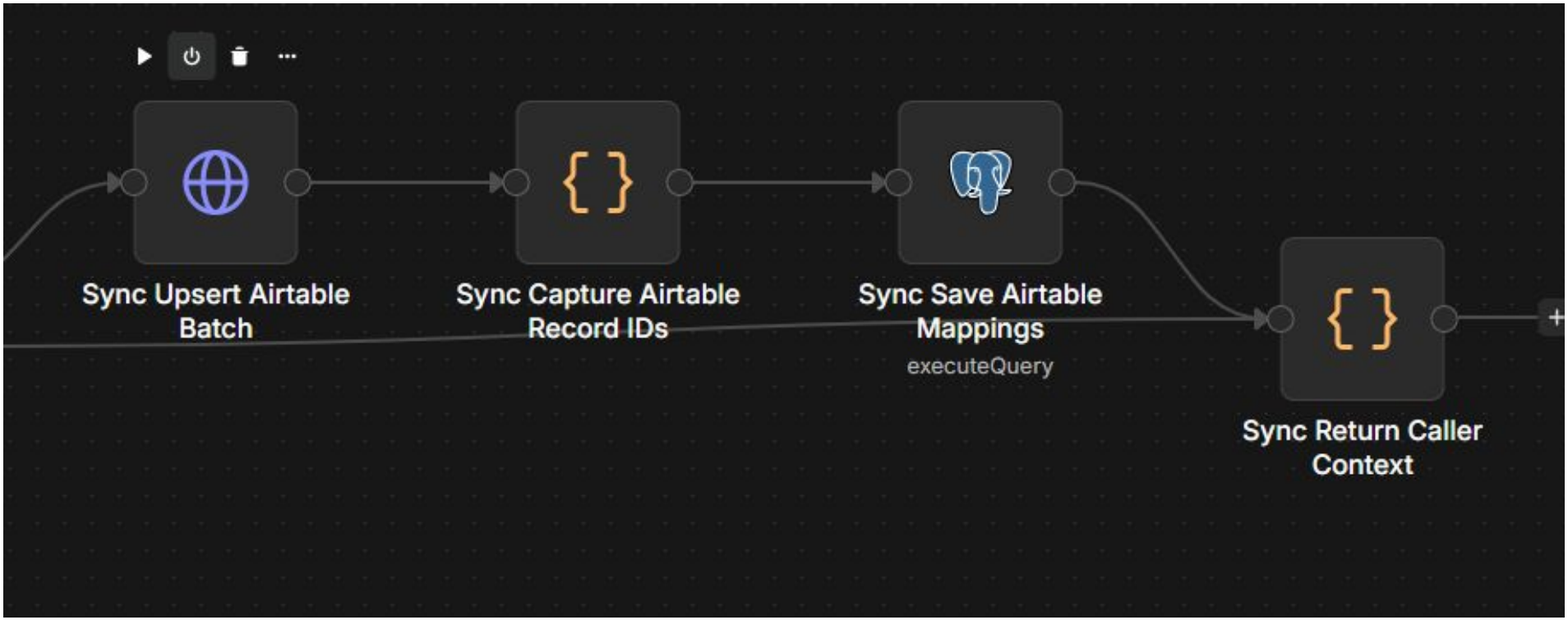


A schedule or state-change call builds a sync run. Changed records are batched for Airtable and sent only when configuration is available. Payload hashes prevent unnecessary updates.

AIRTABLE WORKSPACE / CANVAS CLOSE-UP

Retain record mappings

Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.

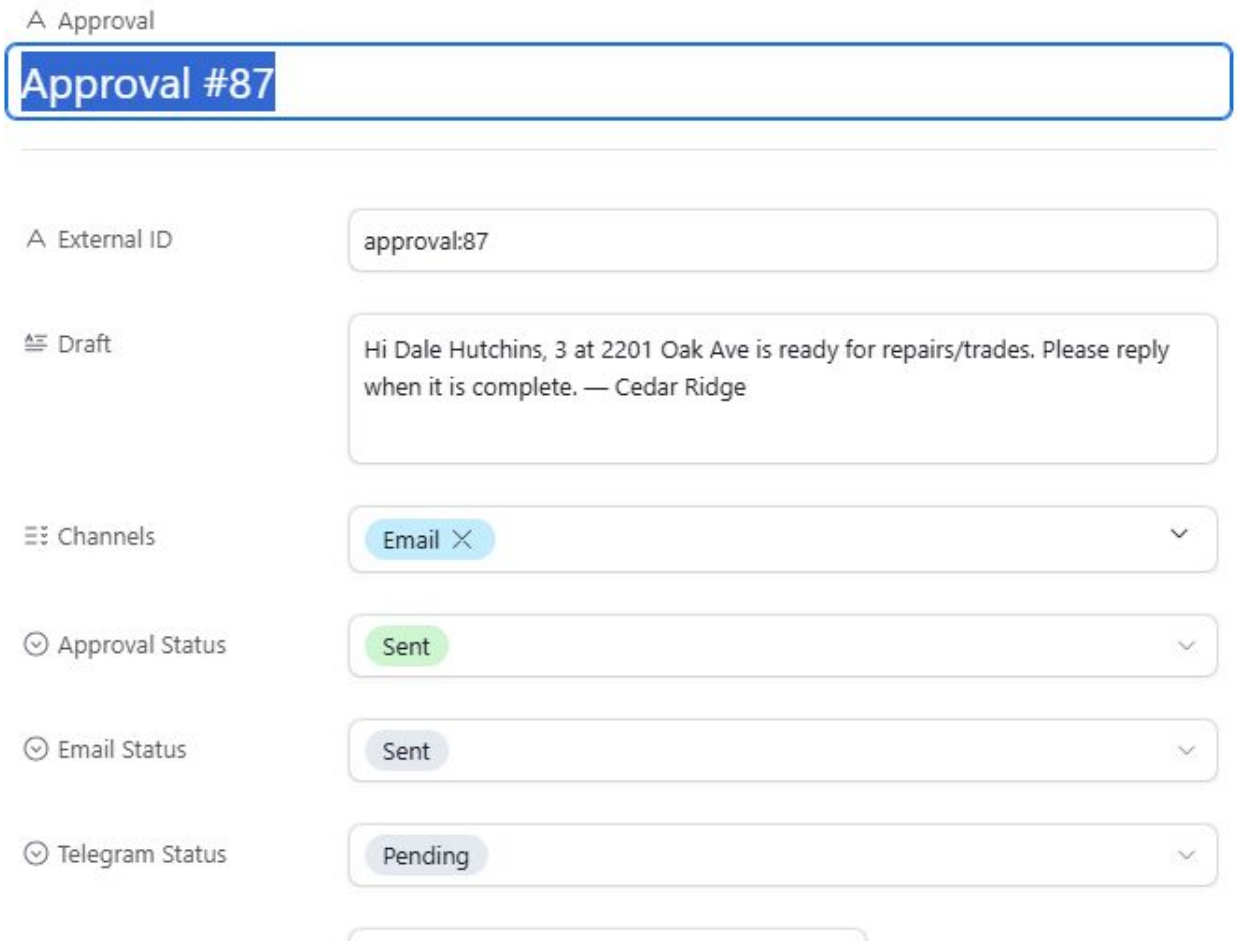


Provider record IDs are paired with their source records and saved in Postgres. These mappings let later updates modify the same mirrored rows and reconcile links between stages, turns and vendors.

AIRTABLE WORKSPACE / OBSERVED RESULT

The repairs email is visibly sent

Actual Airtable record / synthetic Approval 87 / preserved demonstration state

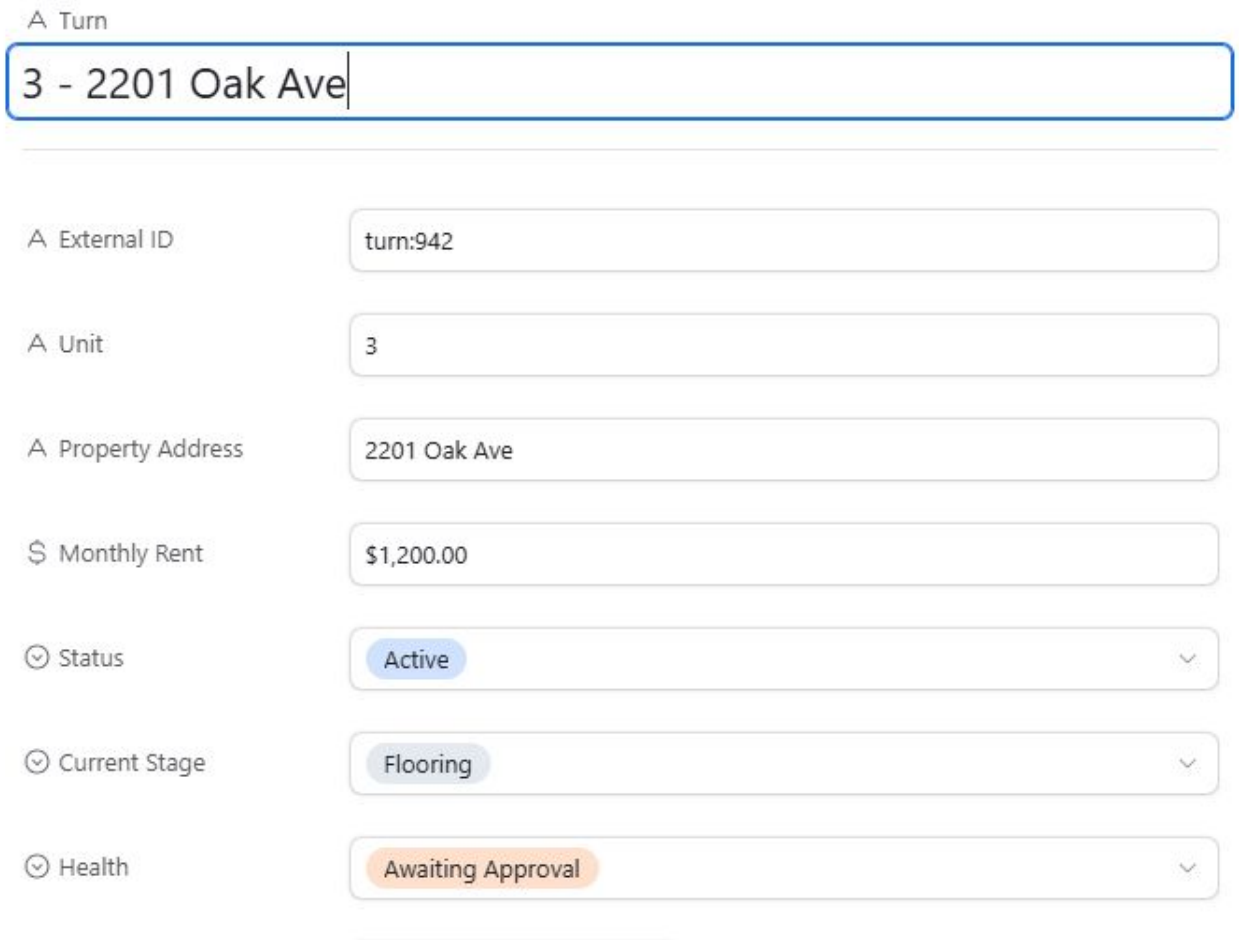


Approval 87 contains the Unit 3 repairs draft, Channels: Email, Approval Status: Sent and Email Status: Sent. Telegram's default Pending value is not a failed attempt: Telegram was not authorized for this handoff.

AIRTABLE WORKSPACE / OBSERVED RESULT

Unit 3 reaches the flooring approval

Actual Airtable turn / fictional Unit 3 at 2201 Oak Ave

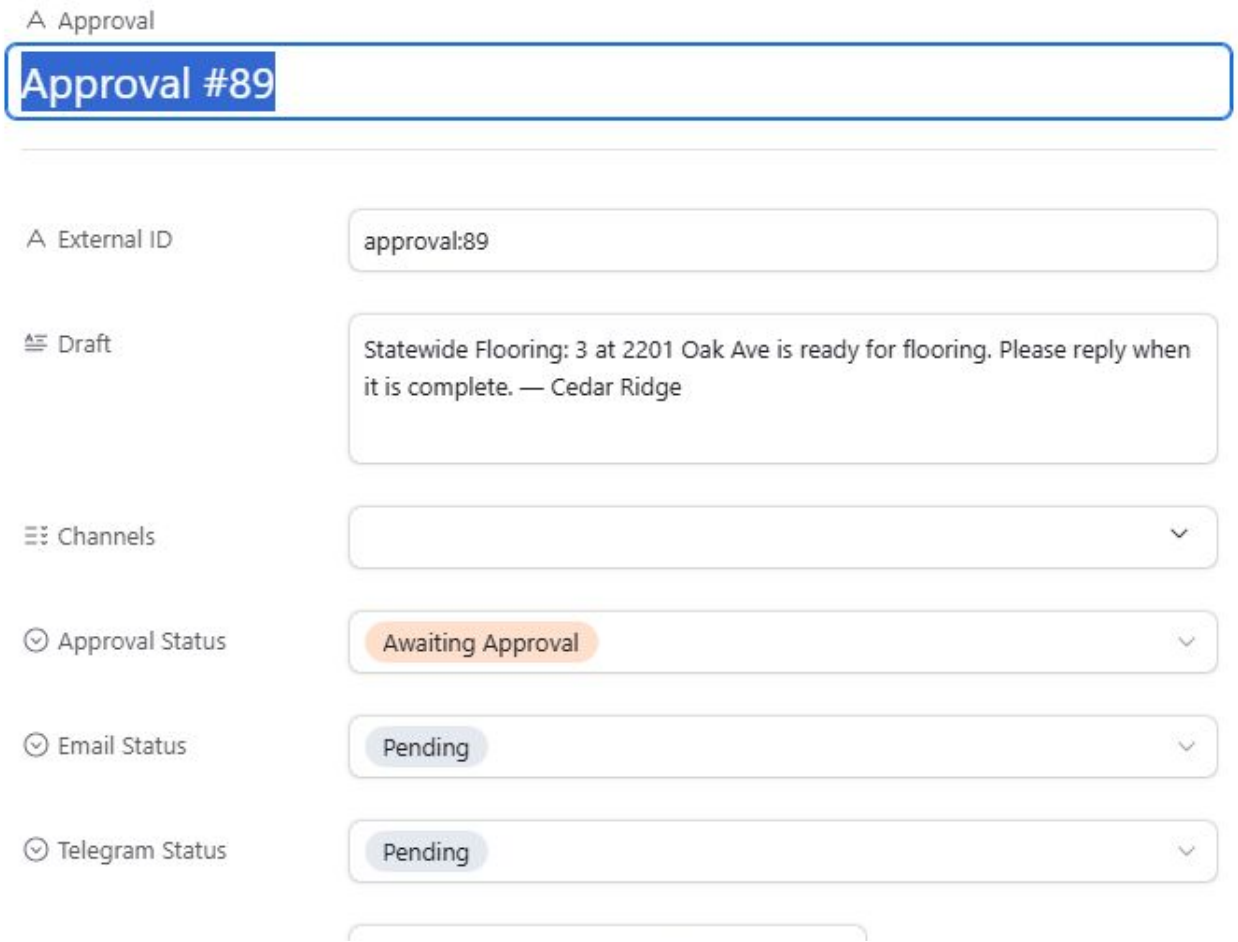


After the repairs and painting replies, turn 942 / Unit 3 is Active with Current Stage: Flooring and Health: Awaiting Approval. This observable record connects the communication journey to the manager-facing operating state.

AIRTABLE WORKSPACE / OBSERVED RESULT

The next message remains unsent

Actual application evidence from the synthetic demonstration.

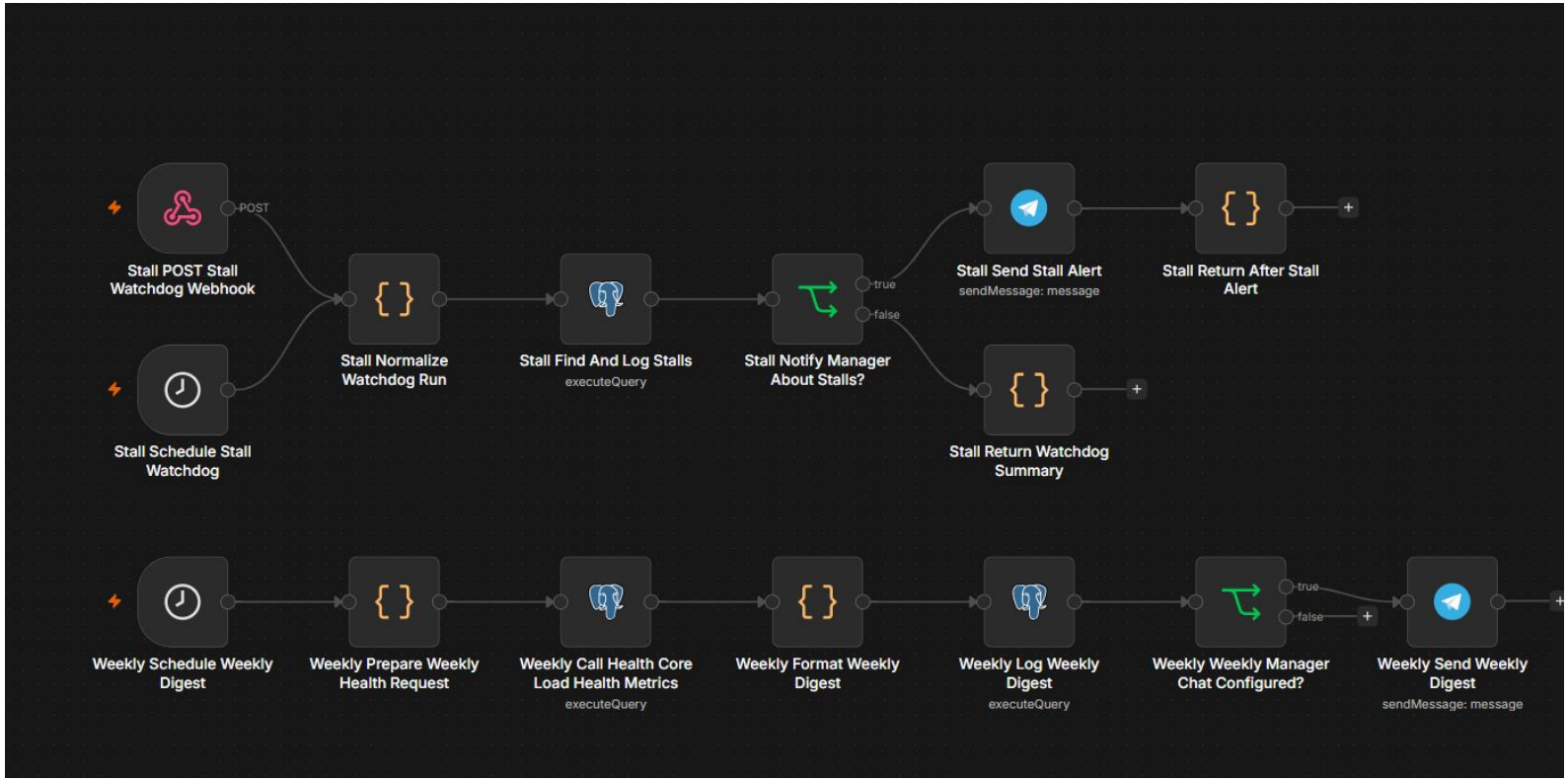


Approval 89 drafts the flooring handoff. Channels is empty, Approval Status is Awaiting Approval, and neither delivery channel is sent. The automation stops at the next manager decision instead of continuing outreach by itself.

WORKFLOW / MONITORING

Monitoring

Scheduled checks turn overdue work into logged events and manager summaries.



COMPLETE CANVAS / 15 NODES / NATIVE ZOOM CLOSE-UPS FOLLOW

TRIGGER / INPUT

Stall schedule or webhook; separate weekly digest schedule.

STEPS / DECISIONS

Read stage age, prior alerts, snoozes and business-hour rules.

DATA / OUTPUT

Log eligible stalls; format the alert or weekly health metrics.

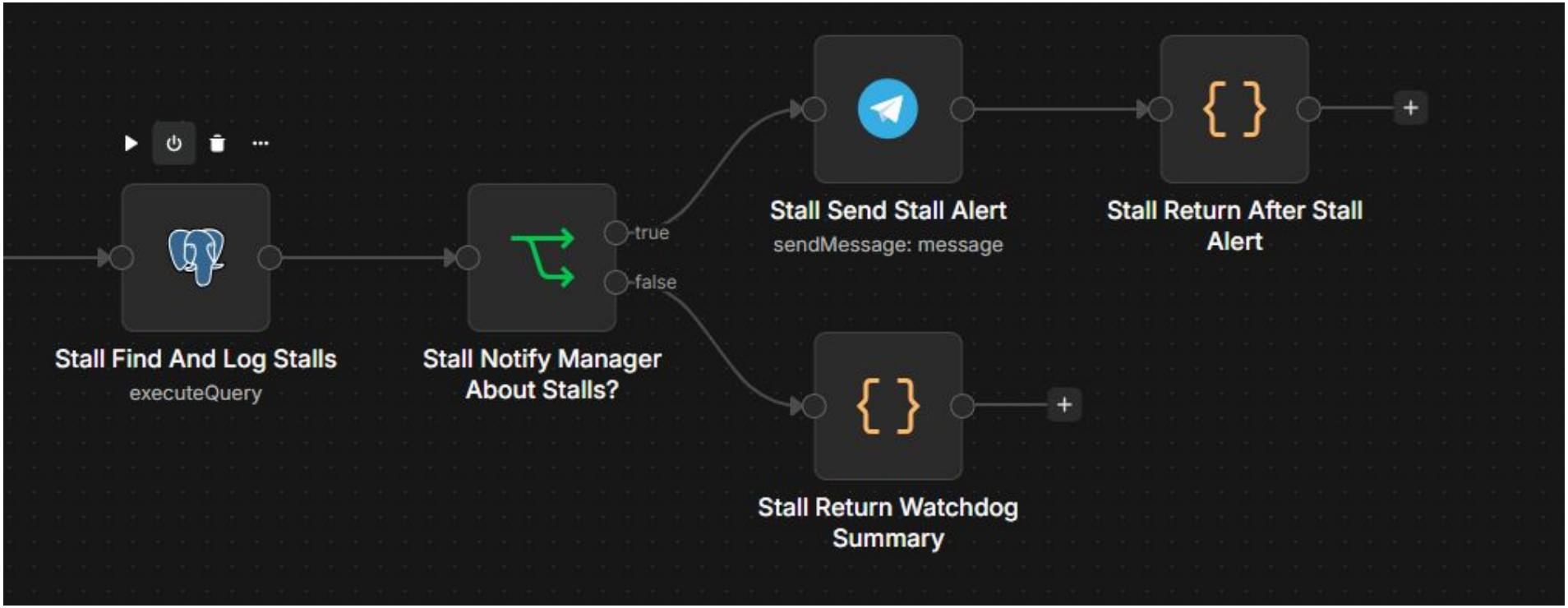
HUMAN CONTROL

Manager receives actionable alerts and can intervene or snooze.

MONITORING / CANVAS CLOSE-UP

Stall detection and notification gate

Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.

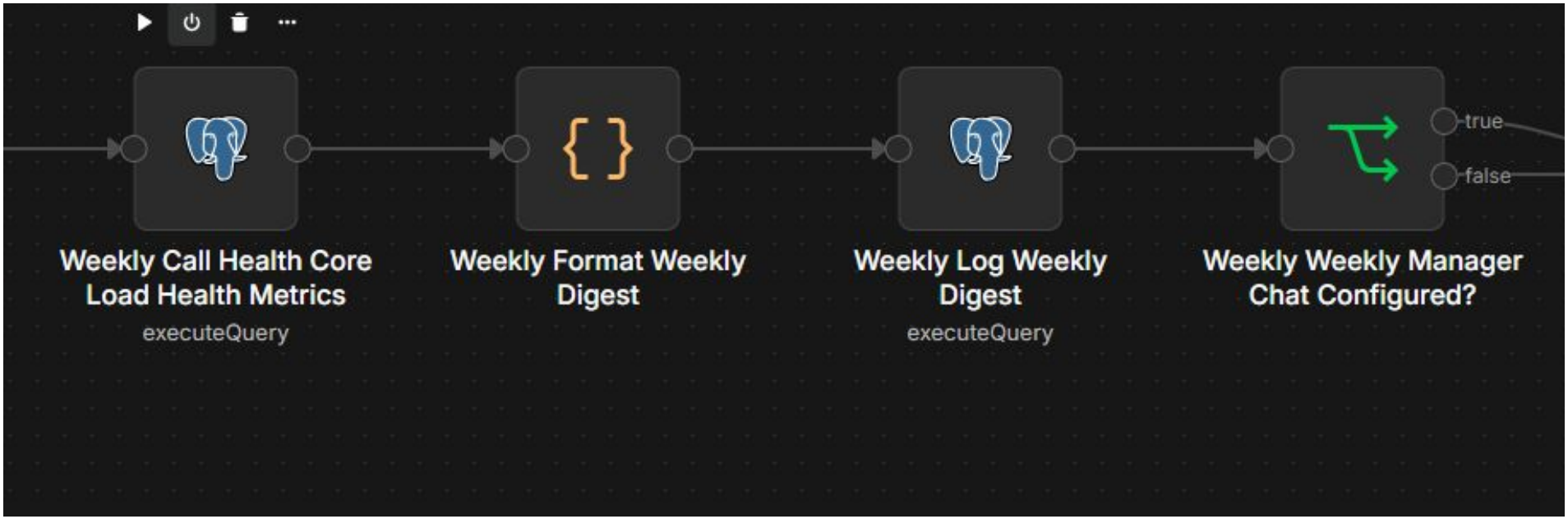


The query compares active stages with their SLA age and previous alert state. The notification gate decides whether the manager bot should send; otherwise the workflow returns the recorded summary.

MONITORING / CANVAS CLOSE-UP

Weekly health digest

Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.

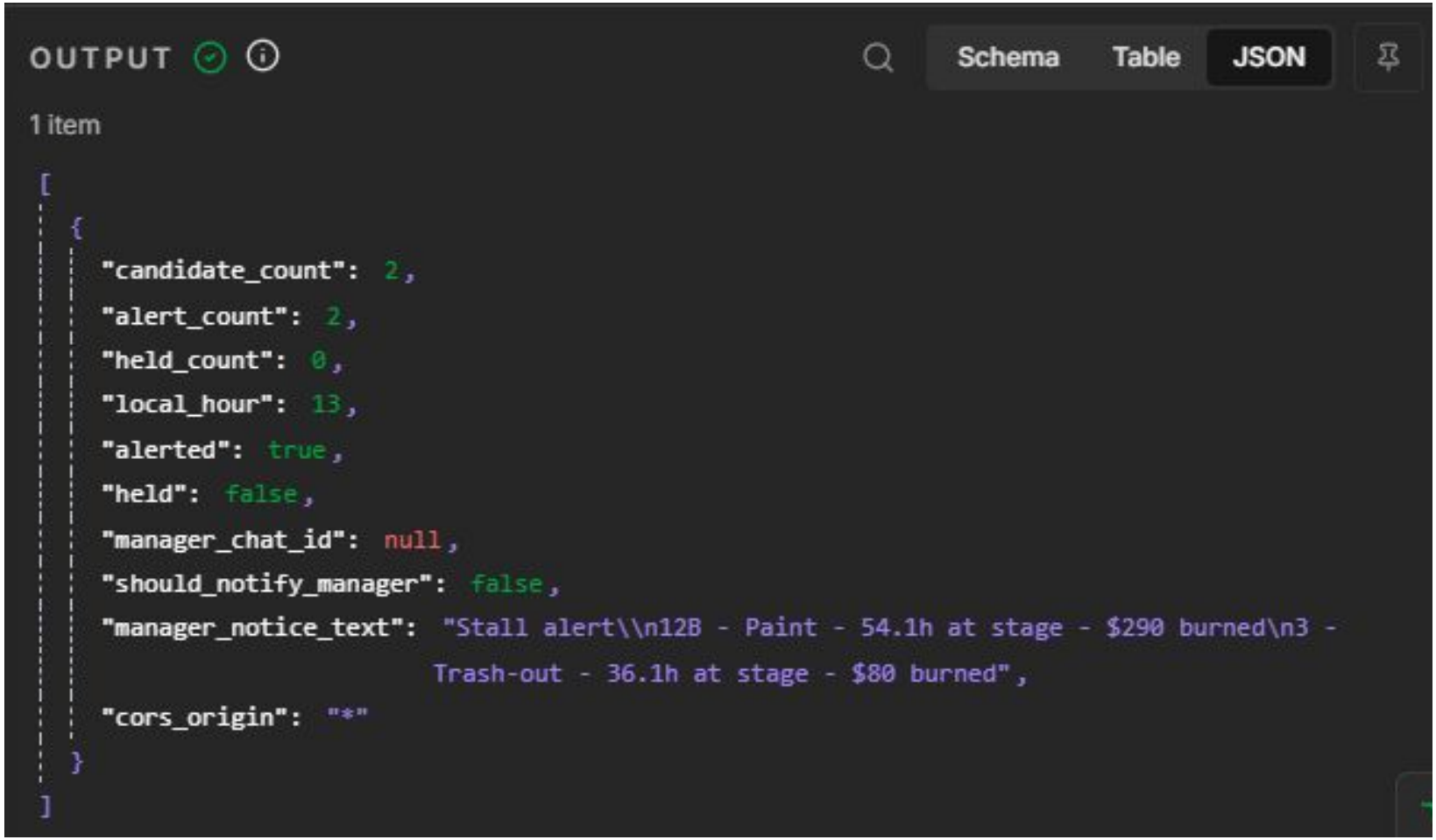


The weekly branch loads stored health metrics, formats a summary, logs it and sends only when a manager chat is configured. The metrics are derived from the automation’s recorded events.

MONITORING / OBSERVED RESULT

Two overdue stages were recorded

Actual application evidence from the synthetic demonstration.

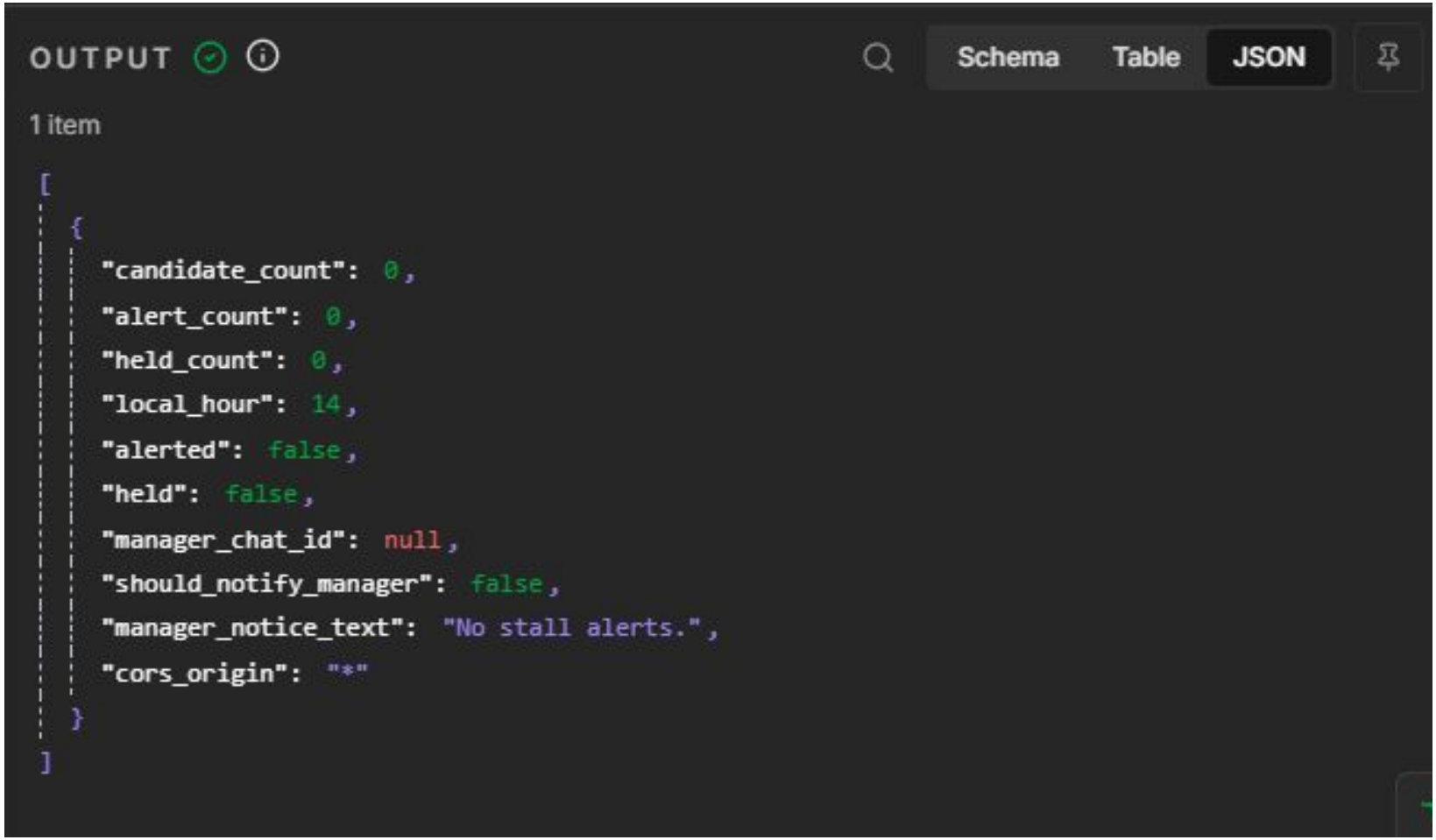


Saved execution 2628 found and logged two eligible stalls: Unit 12B Paint and Unit 3 Trash-out. This run retained the alert text but did not notify Telegram (should_notify_manager: false). The evidence proves detection and logging, without claiming a delivered alert.

MONITORING / OBSERVED RESULT

A later pass has no eligible stalls

Actual application evidence from the synthetic demonstration.

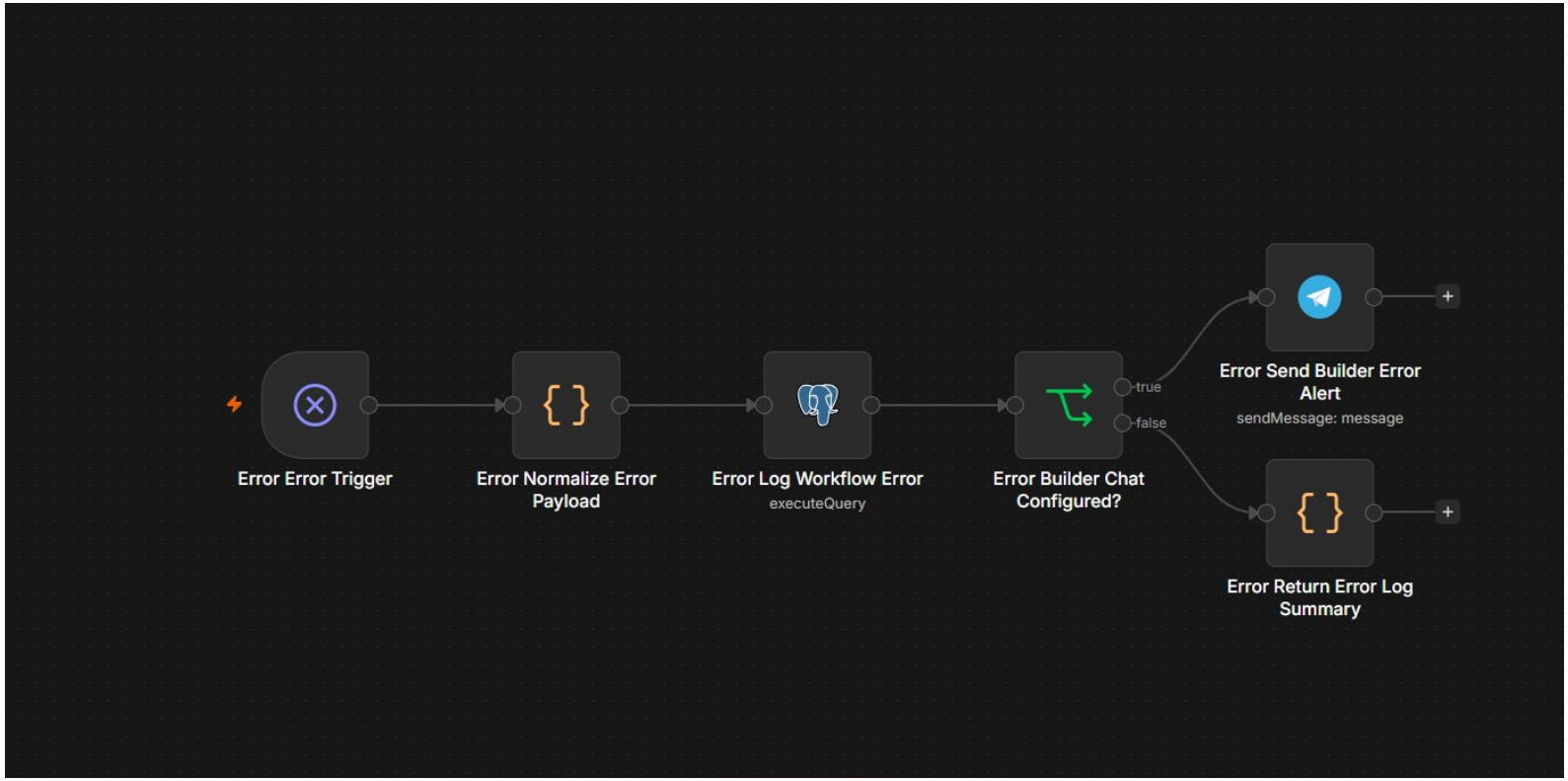


Execution 2687 returned zero candidates and zero alerts. The saved result is an observed later state; it is not presented as a same-state deduplication experiment. The workflow exits without a manager notification.

WORKFLOW / BUILDER ERROR HANDLER

Builder Error Handler

Unexpected execution failures get a durable event and a separate technical notification route.



COMPLETE CANVAS / 6 NODES / NATIVE ZOOM CLOSE-UPS FOLLOW

TRIGGER / INPUT

An n8n error event containing workflow and failed-node context.

STEPS / DECISIONS

Normalize the failure and store a workflow_error event.

DATA / OUTPUT

Return the stored summary; send when the builder route is configured.

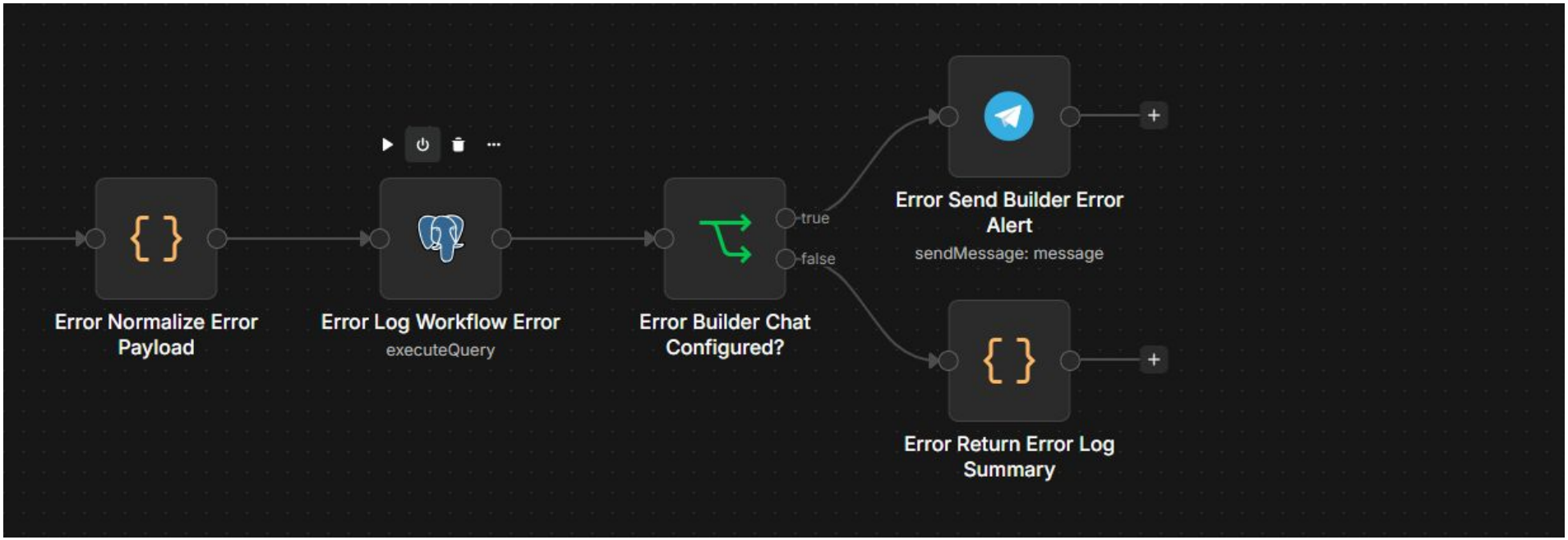
HUMAN CONTROL

A technical operator can investigate without mixing this with vendor messages.

BUILDER ERROR HANDLER / CANVAS CLOSE-UP

Log first, then notify

Actual n8n editor at native zoom. This is a branch view; the complete canvas appears at the start of the section.

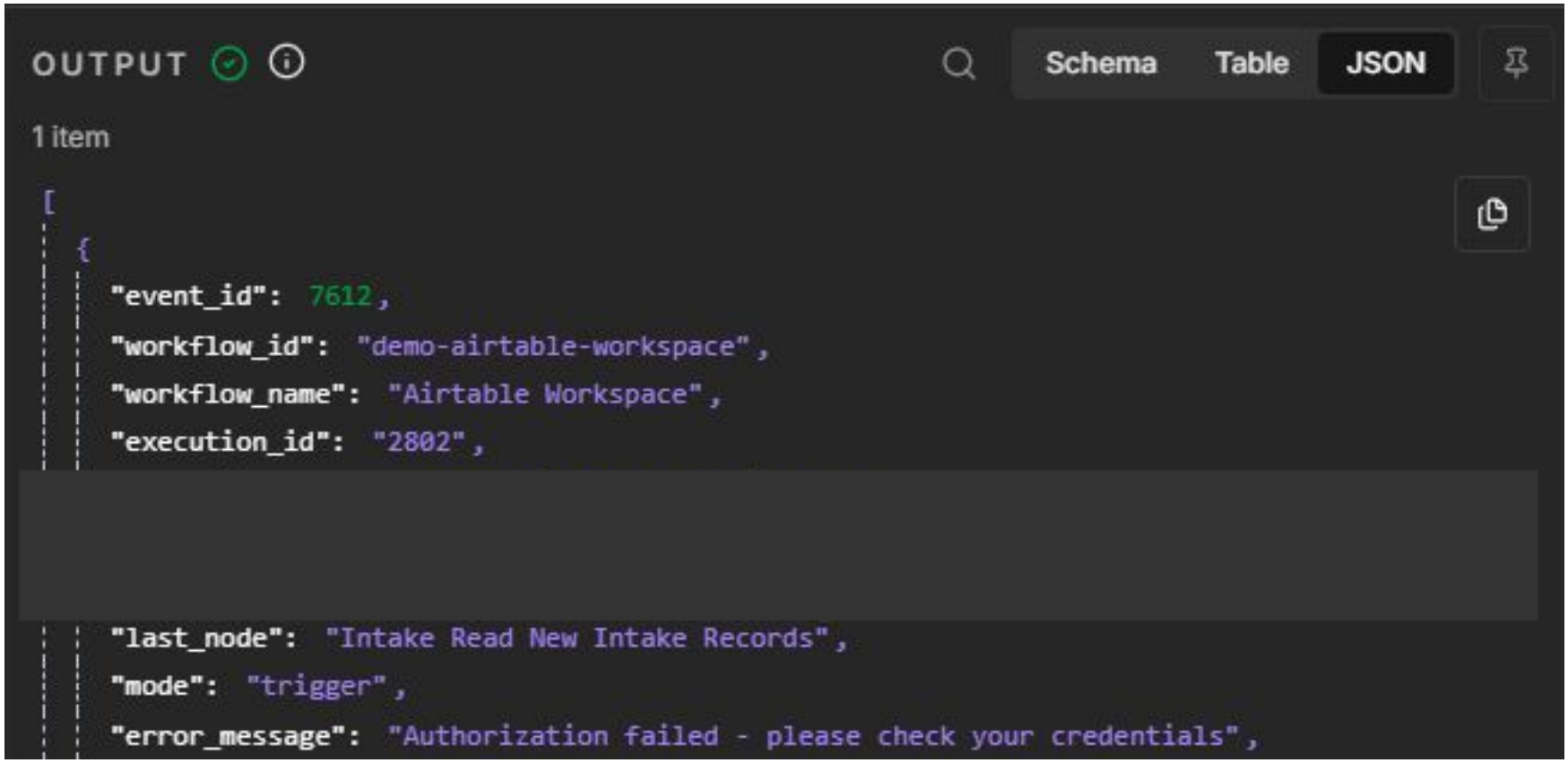


The workflow records the failure before checking the private builder destination. The notification branch and return-summary branch share the same stored event.

BUILDER ERROR HANDLER / OBSERVED RESULT

A real provider failure was logged

Actual error-handler output / October 10 provider-error example



Saved error execution 2803 logged event 7612 for an Airtable authorization failure at the intake read step. The durable row identifies the workflow, failed node and reason. Private execution links and routing identifiers are removed; this view demonstrates error capture.

VERIFIED RESULTS

What the demonstration establishes

Observed coordination behavior with fictional portfolio records.

A CONNECTED MULTI-CHANNEL JOURNEY

Unit 3 moved from a Telegram trash-out reply to an approved repairs email, an approved painting Telegram message and an open flooring approval.

HUMAN APPROVAL TIED TO DELIVERY

Approval 87 has a saved SEND_EMAIL decision, one successful delivery result and an actual Email Sent record in Airtable.

VISIBLE BUSINESS STATE

The vendor conversation shows the painting message received. Airtable shows the same unit at Flooring / Awaiting Approval, with Approval 89 unsent.

CONTROLLED ALTERNATE OUTCOMES

An ambiguous painter reply produced manager_match_needed. A repeated Deep Clean completion was blocked. Stall detection and a provider-error event were recorded.

The earlier guided synthetic scenario rehearsal recorded 20 of 20 passes. The screenshots here add concrete saved executions and observable records to that fixture evidence. No rent uplift, time savings or customer performance rate is claimed.